

A mirror-principle violation meets Nanosyntax: The case of Czech X-marking

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1 Introduction

This chapter addresses the marking of the past tense and the conditional in Czech. In spoken Czech, both forms involve the past participle and the same clitic auxiliary. They differ only in the presence/absence of the conditional clitic *by* ‘would’; e.g., *čet-l* ‘he read’ is the past tense, *čet-l by* ‘he would read’ is the conditional.

The containment between the past and the conditional is not unique to Czech. The literature (including Von Stechow & Iatridou 2023, Majdić 2025) found that such a pattern is also attested in other languages. However, there is a difference between Czech and these other languages: while Czech appears to have the order VERB-PAST COND (*čet-l by*), other languages manifest the order VERB-COND-PAST. This chapter proposes a nanosyntactic account of this paradox, making use of the decomposition proposed in Majdić (2025) and the account of mirror-principle violations explored in Sergienko (2025).

The structure of the chapter is as follows. Section 2 describes the relevant facts and brings out the ordering paradox that arises in Czech compared to other languages. Section 3 provides a Nanosyntactic account of the paradoxical ordering. Section 4 turns to the clitic nature of the conditional and proposes that it starts as an adverbial marker, but moves to the second position. The proposed derivation is used to explain a difference between Czech and Bulgarian regarding the placement of negation in conditional sentences as well as clitic doubling constructions in Czech. Section 5 provides a detailed Nanosyntactic derivation of the full singular paradigm.

2 The puzzle

In this section, I point out a puzzle that arises when the morphology of the Czech conditional is compared to other languages. In Section 2.1, I lay out the morphological facts as found in Czech. In Section 2.2, I provide some comparative data and bring out the paradox.

2.1 *The morphology of the past and the conditional*

The paradigm of the past tense and the conditional in Standard Czech is given in (1). The main point to observe is that the forms always contain a past participle

(in bold) and a clitic auxiliary (which follows the participle in the table). The auxiliary can be segmented into a root and a person/number morpheme that tracks the ϕ features of the subject.

(1) Standard Czech

	SINGULAR		PLURAL	
	PAST	CONDITIONAL	PAST	CONDITIONAL
1	nes-l js-em	nes-l by-ch	nes-l-i js-me	nes-l-i by-chom
2	nes-l js-i	nes-l by-s	nes-l-i js-te	nes-l-i by-ste
3	nes-l	nes-l by	nes-l-i	nes-l-i by

The participle, marked by the suffix *-l*, is also marked for gender/number agreement. In the singular, the masculine agreement is null, but in the plural, we see the masculine animate plural *-i* on the participle. Let us now look in more detail at the clitic forms that follow the participle.

The forms of the past tense auxiliary in Table (1) resemble the forms of the copula ‘be,’ see Table (2) for comparison. The only difference is seen in 3SG and 3PL, where the auxiliary is absent or null.

(2) The auxiliary vs the copula in Standard Czech

	SINGULAR		PLURAL	
	AUXILIARY	COPULA	AUXILIARY	COPULA
1	js-em	js-em	js-me	js-me
2	js-i	js-i	js-te	js-te
3	Ø	je	Ø	js-ou

In Standard Czech, the subjunctive forms also have a zero person marker in the 3RD person. Thus (in the 3RD person), the conditional differs from the past tense only by the presence/absence of *by*. In the other persons, also the person markers are different, and the difference between the forms is thus more complex than the pure presence/absence of *by*.

Interestingly, in spoken Czech, the ϕ agreement following the conditional *by* becomes identical to the past tense, so that the two forms differ only by the presence/absence of *by* in all persons. This is shown in Table (3).¹

(3) Paradigm levelling in the spoken language

	SINGULAR			PLURAL		
	PAST		CONDITIONAL	PAST		CONDITIONAL
1	nes-l	s-em	nes-l by s-em	nes-l-i	s-me	nes-l-i by s-me
2	nes-l	s (s-i)	nes-l by s (s-i)	nes-l-i	s-te	nes-l-i by s-te
3	nes-l		nes-l by	nes-l-i		nes-l-i by

Assuming that the auxiliary in the past tense and in the conditional simply track the ϕ features of the subject, we can posit the template in (4).

¹Different dialects mix the Standard forms with the spoken forms in different proportions, see, e.g., Sgall & Hronek (2014:48-9). In the 2ND person, two different forms of the auxiliary exist.

- (4) The past and the conditional
- | | | | |
|----|-------------|----------------------|----------|
| a. | PAST | Past participle | + ϕ |
| b. | CONDITIONAL | Past participle + by | + ϕ |

The template (4) brings out, in an abstract format, the fact seen in Table (3), which is that the past tense and the conditional differ by the presence/absence of *by* in all persons.

The next section brings out an apparent paradox that has to do with the ordering of the suffix of the past participle (*-l*) and the conditional *by*. The paradox stems from cross-linguistic comparison. In particular, I will show that while other languages show a similar containment pattern where the conditional adds a morpheme compared to the past tense, they show a different order of the past-tense marker and the conditional marker, with the conditional closer to the root than the past tense.

2.2 Crosslinguistic comparison and the puzzle

One of the uses of the Czech conditional is in contexts that Von Fintel & Iatridou (2023) call X-marking. This marking is used, among others, in counterfactual antecedents of conditionals, as in (5a). In the sentence, it is implied that the antecedent is not true (the relevant person is absent). The indicative in (5b) simply expresses an open possibility (e.g., I am in a large auditorium and I am unaware if the relevant person is present).

- (5) a. Pokud **by** tu (ted') **by-l**, **by-l** **by** šťastný.
 if COND here now be-PAST be-PAST COND happy
 ‘If he were here (now), he would be happy.’
- b. Pokud tu (ted') je, je šťastný.
 if here now be.PRES.3SG be.PRES.3SG happy
 ‘If he is here now, he is happy.’

Von Fintel & Iatridou (2023) use the term antecedent X-marking to refer to the marking found in the antecedent clause in (5a). They use this term to avoid ambiguous use of terminology in the literature, where these forms are sometimes referred to as conditional, counterfactual, or subjunctive. They observe that X-marking is also frequently used in the consequent of the counterfactual conditional and in *wish* clauses (both in the matrix and embedded clause), as in (6):

- (6) Přá-l **by** si, a-**by** odje-l-i.
 wish-PAST COND self.DAT that-COND leave-PAST-PL
 ‘He wishes that they left.’

Von Fintel & Iatridou (2023) point out that even though X-marking constructions are not ‘past’ in any clear sense (note the adverb ‘now’ in (5a)), the fact that past tense morphology is used in these constructions is quite frequent across languages. In some languages, X-marking is even identical to the past tense, see (7b), as an example.

- (7) a. If Miranda knows the answer, Emily knows the answer.

- b. If Miranda **knew** the answer, Emily would know the answer.

Majdić (2025) further observes that the Czech situation, where X-marking contains past tense plus some other marker, is relatively frequent. This is the case, for instance, in Turkish. The Turkish past tense marker *-(y)di* (seen in (8a)) is preceded by the conditional *-se* in counterfactuals such as (8b), and also in *wish* clauses, see (8c).

(8) Turkish (Kornfilt 2013:243, Von Stechow & Iatridou 2023:1488)

- a. Hasan Ankara-dan gel-**di**.
Hasan Ankara-ABL come-PAST
'Hasan came from Ankara.'
- b. John önümüzdeki salı gel-**se-ydi** annesi çok mutlu
John next Tuesday come-SA-PST his.mom very happy
or-ur-du.
be(come)-AOR-PST
'If John arrived next Tuesday, his mom would be very happy.'
- c. Keşke önümüzdeki salı gel-**se-ydi**.
keşke next Tuesday come-SA-PST
'I wish he would come next Tuesday.'

Another example of this phenomenon that Majdić (2025) found comes from Mauwake (a Nuclear Trans New Guinea language). The past tense *-a* is shown in (9a). In *wish* clauses and counterfactual conditionals, the past tense *-a* is preceded by the 'counterfactual' *-ek*, see (9b,c) respectively. Note that in the Mauwake counterfactual, X-marking is found both in the antecedent and the consequent, similarly to what we see with Czech counterfactual conditionals, recall (5a).

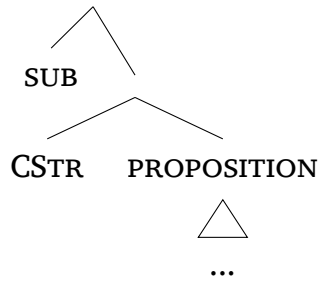
(9) Mauwake (Berghäll 2015:157, 377, 367, via Majdić 2025)

- a. miim-**a-k**
hear-PAST-3SG
'He heard.'
- b. [Yo = ko wia uruf-**ek-a-m**] na-ep.
1s.unm = nf 3P.ACC see-CNTF-PAST-1S say-SS.SEQ
'I would like to see him.'
- c. [Yena aamun aakisa uruf-**ek-a-m** = na]
1S.GEN S/P.younger.sibling now see-CNTF-PAST-1S = TP
kemel-**ek-a-m**.
be.happy-CNTF-PAST-1S
'If I saw my younger brother now, I would be happy.'

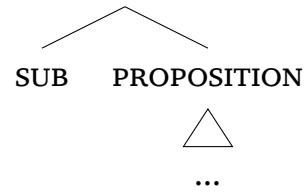
Taking stock, there is a commonality that Czech and the other two languages share, which is that X-marking constructions combine the past tense morphology and an extra (conditional/irrealis) marker. Majdić (2025) provides a semantic account of X-marking that builds on this observation and proposes that the semantics of X-marking has at least two ingredients, namely the past tense (which he calls SUB) and an additional head, which he calls CSTR, see (10).

(10) The decomposition of X-marking in Majdić (2025):

a. X-marking



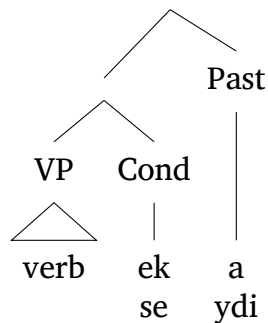
b. Past



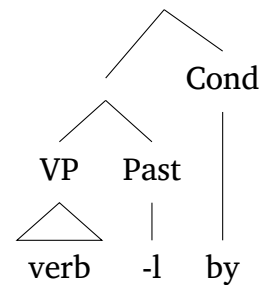
The semantics of the ingredients is set up in such a way that when the SUB feature applies to the proposition directly, as in (10b), the result is a temporal back shift (i.e., past tense). When it applies to the constituent headed by CSTR, as in (10a), the result is the semantics that we find in X-marking constructions.

Taking this for granted, what appears to be puzzling is the apparent contrast in the order of morphemes between the languages discussed here vs Czech. Specifically, in Turkish and Mauwake, the conditional marker (analogous to *by*) is closer to the verb than the past tense marker, see (11a). In Czech, the ordering is different and suggests a different hierarchy of the projections, see (11b).

(11) a. Mauwake, Turkish



b. Czech



The paradox is not merely linear but pertains to the status of the morphemes. Specifically, the Czech ‘past’ *-l* is a suffix, and it is always glued to the verb. On the other hand, the conditional marker is a clitic and obligatorily moves to the second position. This is shown in (12), which is a single conditional with the antecedent in (12a) and the consequent in (12b). The antecedent has the complementiser *pokud* in the clause-initial position, and the clitic must follow it immediately. In the consequent (see (12b)), the clitic remains after the verb, since the verb is first in the clause, showing the mirror-violating order.

- (12) a. *Pokud by dnes přišel Marcin,*
 if COND today come-PAST.PART Marcin,
 b. *přišel by i Radim.*
 come-PAST.PART COND also Radim.
 ‘If Marcin came today, Radim would come too.’

Thus, the issue with Czech is not only the linear arrangement, as in (12b), but also the fact that the conditional is obviously only loosely associated to the verb,

which again suggests that *-l* is closer to the verb (in Czech) than the conditional, as depicted in (11b).

The independent status of the conditional (as opposed to the bound nature of the *-l* in the past participle) stands out when we compare it to Balkar (Kasenov 2022). In this language, the conditional is expressed by *se*, seen in (13a). In the X-marking construction (i.e., in the counterfactual conditional), the *se* is accompanied by the past tense, see (13b). Unlike what we saw in Czech (12a), it is the past tense that is expressed as an auxiliary, and the conditional is suffixed on the main verb.

(13) Balkar (Kasenov 2022)

- a. men šorpa et-**se**-m, [Kerim quan-iriq-di]
 1SG soup make-COND-1SG Kerim happy-FUT-3SG
 ‘If I make a soup, Kerim will be happy.’
- b. men šorpa et-**se** e-**di**-m, [Kerim quan-iriq e-di]
 1SG soup make-COND AUX-PST-1SG Kerim happy-FUT AUX-PST
 ‘If I made a soup, Kerim would be happy’

The mirror-violating nature of the Czech conditional thus only becomes even more apparent, even taking its clitic status into consideration. The point is that it’s precisely its clitic status (as opposed to the bound nature of the past participle *-l*) that reinforces the paradox.

3 A Nanosyntax account of the Czech paradox

This section proposes a solution to the apparent mirror-principle violation using the framework of Nanosyntax (see, e.g., De Clercq et al. 2025).

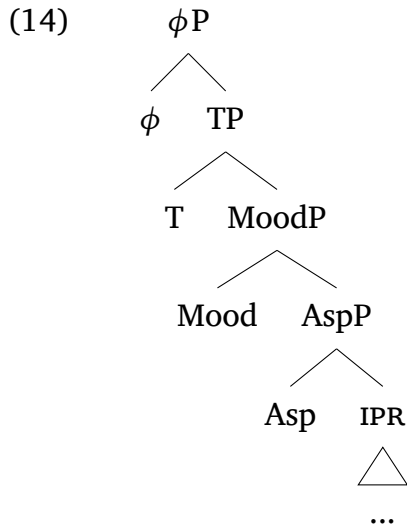
The framework has two core assumptions. The first one is that natural language has only a single grouping operation that forms complex structures from simplex ingredients, and that is (binary) merge. All complex objects are assembled by this operation. This entails that there are no pre-syntactic feature bundles, since feature bundles are complex objects created out of features. As a result, all such objects must be created by merge and have the regular binary structure (Starke 2013). The second core assumption is that the features of languages are drawn from the universal set of features dedicated to language (as in, e.g., Chomsky 2004, Cinque 2013). The first step of any analysis thus requires one to specify the relevant features and their hierarchy. This is the goal of Section 3.1.

The second step of an analysis is to determine how these structures are lexicalised, using language-specific lexical items. Section 3.2 makes explicit the basic ideas underlying the process of lexicalisation in Nanosyntax, illustrating it’s working on the morphological relations between X-marking and past tense morphology in German. Section 3.3 show how the ordering paradox is resolved, drawing on the ideas explored in Sergienko (2025).

3.1 The syntactic structures

This section presents the underlying syntactic structures. A course-grained approximation of the structure I assume is provided in (14). The tree has the ‘theta

domain’ at the very bottom, marked in (14) by the abbreviation IPR. The individual letters stand for Initiation, Process and Result, the three heads that make up the theta domain in Ramchand (2008).



Above the theta domain, we find the projections responsible for Aspect, Mood and Tense. Finally, at the top of the tree, I assume the presence of agreement features, which add presupposition regarding the ϕ features of the subject.

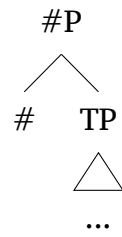
I adopt this hierarchy based on the existing Nanosyntax literature (Starke 2020, Starke & Cortiula 2021, Cortiula 2025, Vanden Wyngaerd 2025) and I will argue that it allows us to explain the facts reported in Section 2. Specifically, I will analyse the conditional markers found in X-marking constructions as lexicalising Mood, which fits the evidence from Turkish and Mauwake where these markers appear in between the verb and the tense marker (leaving the issue of the Czech order aside for now).²

Let me now turn to further distinctions within the ϕ , Tense, Mood and Aspect domains. Since Nanosyntax rejects feature bundles and works only with privative features, various sub-distinctions within these categories must be made by adding more features into the functional sequence. For ϕ features, I assume that number is located below person and that plural is marked with respect to singular by the presence of PL. Singular is therefore the default interpretation of the number node (#), see (15a), with plural added on top, see (15b).³

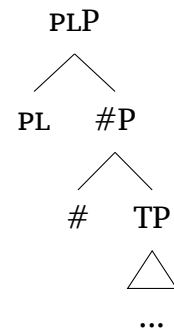
²The hierarchy (14) differs from the one proposed by Cinque (2014), where Mood is higher than Tense. It is possible that this boils down to terminology, since it’s not clear to what extent the X-marking morphemes discussed here fall under the same set of markers that Cinque labels Mood, as he mainly studies declarative and interrogative markers. Note also that the X-markers under investigation are compatible with complementisers in the C-domain (recall (12), (7)), which is another reason for placing them lower down in the structure.

³For reasons of space, I will only discuss the singular paradigm, so the PL feature will not be used in what follows.

(15) a. singular

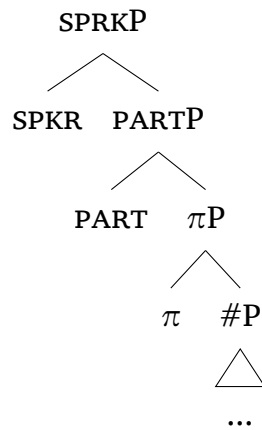


b. plural

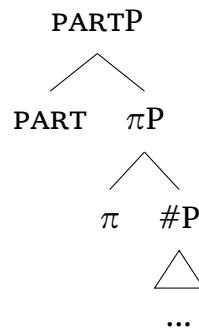


On top of number, π denotes the default person (which the 3RD person), see (16c). The 2ND person is a default interpretation of the PARTICIPANT feature, see (16b). The 1ST person has SPEAKER in addition, see (16a).

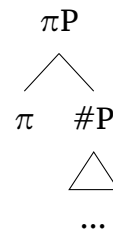
(16) a. 1ST person



b. 2ND person

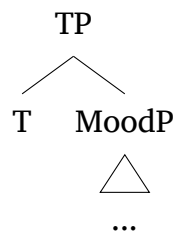


c. 3RD person

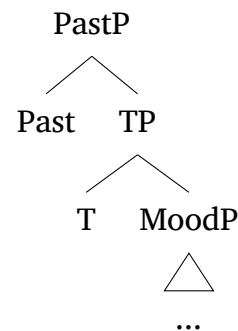


For Tense, I follow the work cited and analyse the present tense as a plain T head, as in (17a), while the Past tense adds Past on top of T, as in (17b).

(17) a. present tense



b. past tense



Finally, let me turn to Mood. The current work in Nanosyntax mainly looked at the subjunctive and the indicative moods. In the existing literature, the subjunctive is considered unmarked and the indicative marked; see Starke & Cortiula (2021), Vanden Wyngaerd (2025). However, despite the fact that there is some overlap between the conditional (investigated here) and the subjunctive, they are not the same thing. To see that, consider French, as discussed in Von Fintel & Iatridou (2023). The authors point out that French has a past tense (see (18a))

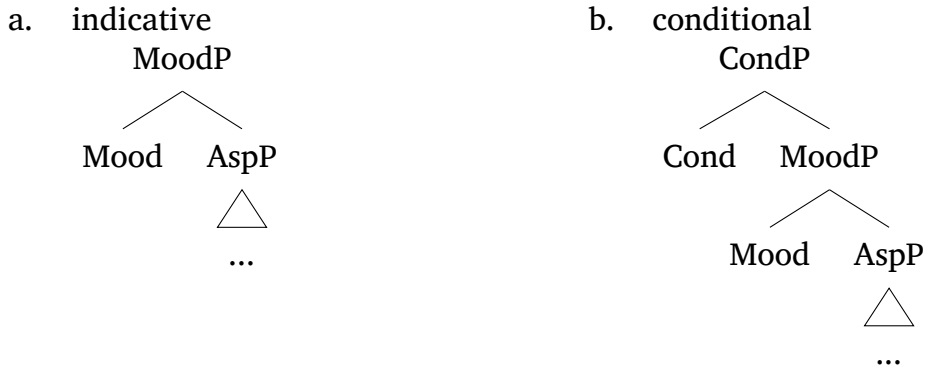
clearly distinct from a subjunctive (see (18b)). Yet X-marking uses the past and not the subjunctive, see (18c).

(18) French (Von Fintel & Iatridou 2023)

- a. Marie **avait** un parapluie rouge hier.
 Marie have.PST.IND a umbrella red yesterday
 ‘Marie had a red umbrella yesterday.’
- b. Je doute que Marie {**ait** / *avait} hier un
 I doubt that Marie have.SUBJ have.PST.IND yesterday a
 parapluie rouge.
 umbrella red
 ‘I doubt that Marie had a red umbrella yesterday.’
- c. Si Marie {**avait** / *ait} un parapluie rouge, il
 if Marie have.PST have.SUBJ a umbrella red, he
 {l’aurait / *ait} vu.
 it.have.COND have.SUBJ seen
 ‘If Marie had a red umbrella, he would have seen it.’

Therefore, I will consider the conditional a distinct Mood from the subjunctive. Following the proposal by Majdić (2025), I shall treat the indicative as unmarked relative to the conditional, see (19a). Under this account, the conditional has an extra meaning ingredient, which I refer to as Cond in (19b), and take it as identical to Majdić’s CSTR.⁴

(19) The structure of the indicative and the conditional

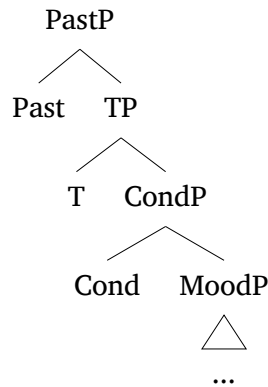


With the decomposition in place, recall that for Majdić (2025), X-marking semantics consists in projecting the past tense on top of the conditional feature, recall (10). Within the current decompositional approach, the structure of X-marking is as depicted in (20a), and the structure of the past indicative as in (20b).

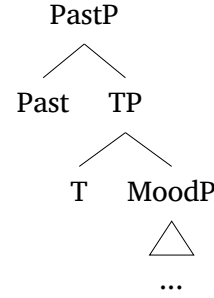
⁴I leave it open whether it is possible to formulate an account where the indicative has an extra feature compared to the conditional.

(20) The structure of the past and X-marking

a. X-marking



b. Past indicative



3.2 Lexicalisation

To illustrate how lexicalisation works, I shall briefly discuss the realisation of the past tense and X-marking in German. The relevant cases I will consider are provided below.

(21) The forms of the past and X-marking (Majdić 2025)

	Present Ind.	Past	X-marked
feel	du fühl(-st)	du fühl-te(-st)	du fühl-te(-st)
need	du brauch(-st)	du brauch-te(-st)	du bräuch-te(-st)
go	du geh(-st)	du ging(-st)	du ging-e(-st)
sing	du sing(-st)	du sang(-st)	du säng-e(-st)

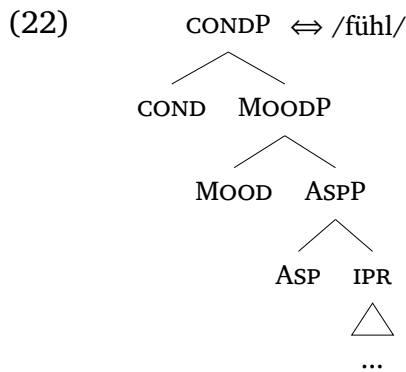
On the first line, there is syncretism between the past tense and X-marking (*fühl-te*; -st is the second person singular marker). Syncretism between the X-marked form and past is what we saw already in English, recall (7). On the second line, the X-marked form has the same suffix as the past form (-te), but a different vowel in the root (*brauch* ~ *bräuch*). If the vocalic alternation (ablaut) were to be considered as triggered by an independent marker, these cases would resemble Turkish in that X-marking has an additional morpheme (ABLAUT) that appears in between the root and the past ([ROOT-ABLAUT]-PAST). On the third line, we are looking at a case that is similar to Czech, because the X-marked form is apparently based on the past tense (*ging-*) by adding a morpheme dedicated to the X-marked form (*ging-e*). Finally, on the last line, the X-marked form has both a special affix and a dedicated root shape.

In sum, the table illustrates that within a single language, German, we find very much the full range of options how X-marking is realised across the different languages represented in this chapter. This suggests that any morphological theory must be able to derive this range of realisations from a single underlying morphosyntactic representation even within a single language. Once such a theory is available, it seems reasonable to extend this theory to subsume also cross-linguistic variation: what we want is a theory of X-marking that assumes a universal underlying meaning for all languages (an assumption this chapter shares

with Von Fintel & Iatridou 2023 and Majdić 2025), and where all variation is a reflex of different lexical items found in the different languages.

To achieve this goal, let me now introduce the principles governing the realisation of syntactic structure within Nanosyntax. The first core idea is phrasal lexicalisation. Phrasal lexicalisation is linked to the idea that every feature is a separate terminal. If that is so, then morphemes – which generally lexicalise nontrivial sets of features – must be able to lexicalise multiple terminals. Starke’s (2009) idea is that when lexicalisation targets a set of features (corresponding to multiple terminals), it targets sets that are independently provided by syntax – constituents. This approach is theoretically simplest in that no new grouping mechanism other than merge is needed, and it is also the same approach as used for other phenomena such as movement: when multiple terminals move together, we say that this is because they form a constituent, and it is the constituent that moves. Similarly, when multiple terminals undergo lexicalisation together, this is because a constituent containing these terminals undergoes lexicalisation.

Suppose now that the phonology of the German root ‘feel’ is stored in the lexicon as linked to a full phrase containing all the verbal features up to and including COND. Such a lexical item is shown (in a simplified fashion) in (22).



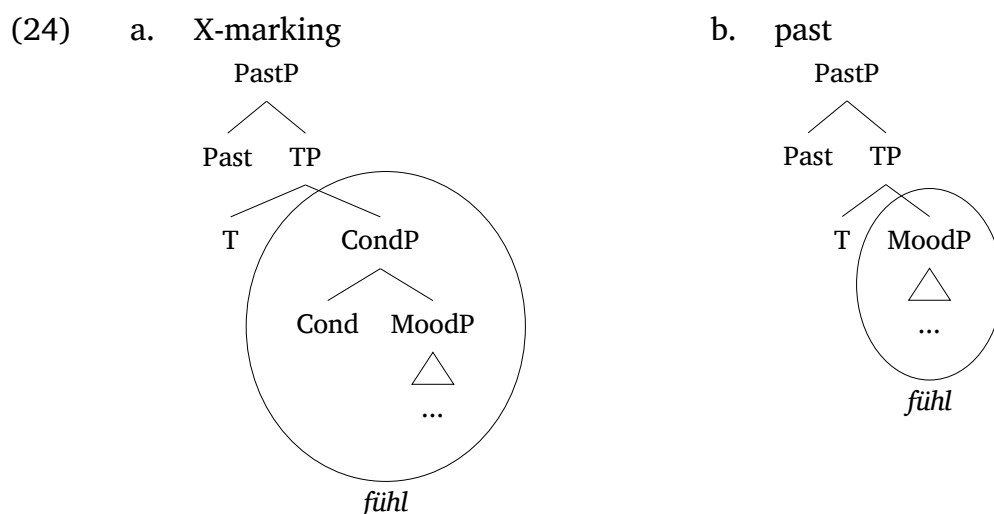
The lexical item is nothing but a stored link between a well-formed representation in the syntax module and a well-formed phonological representation. Its use during sentence construction is such that when syntax produces a tree that is identical to the stored lexical tree, the syntactic representation can be mapped onto the PF representation *fuehl*. The principle that determines what lexical item can map a particular syntactic structure onto PF is the so-called Matching Condition, and it is given in (23).

- (23) *Condition on matching* (De Clercq et al. 2025)
A lexically stored constituent L matches a syntactic phrase S iff S is identical to L.

Using this condition, there are multiple syntactic representations that can be mapped onto the phonology *fuehl*. That is because not only the full tree in (22) is a ‘lexically stored constituent,’ any subconstituent of (22) is also a constituent that is stored in the lexicon. If the topmost CondP is stored, by necessity, all subconstituents are stored as a part of it. Therefore, any subconstituent can be lexicalised as *fuehl*.

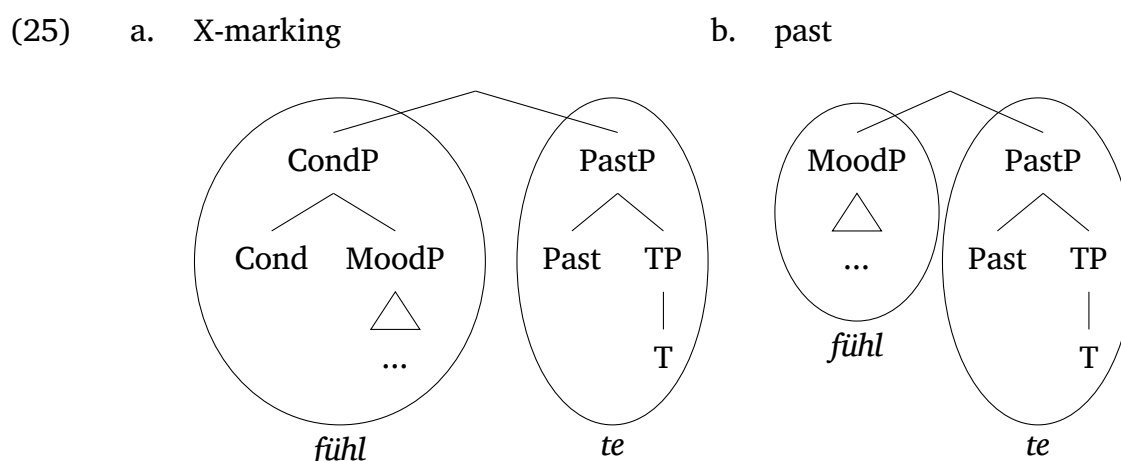
To see how this works, suppose we have the structures for X-marking and

for the past tense as in (24). In the morphosyntactic structure of the X-marking construction, there is a constituent – CondP – that is identical to the whole lexically stored constituent in (22). Therefore, this constituent can be lexicalised by mapped onto *fühl* at PF.



In the past tense, see (24b), there is no CondP, but there is a MoodP constituent that is stored as a part of the lexical entry for *fühl*. Therefore, there is again an identity between the syntactic constituent circled in (24b) and a constituent stored in the lexicon. Therefore, MoodP is lexicalised as *fühl*.

Now notice that after the root lexicalises the maximum structure it matches, the root leaves an identical ‘residue’ for lexicalisation by other items. In cases like these (as I shall discuss in short), the lexicalisation procedure evacuates the constituent lexicalised by the root out of the PastP, leading to the tree representations depicted in (25). In these trees, the features T and Past form a constituent and the morpheme *-te* lexicalises both features in each of the forms.



Thus, as a result of how matching works, the form used for X-marking and the one for past tense end up syncretic for the lexical item *fühl*, as the pieces lexicalising the two structures in (25) are identical. The syncretism is due to the fact that the root lexicalises two different structures, one subpart of the other, which is a consequence of the Matching Condition (23).

For simplicity, the result of lexicalisation can also be depicted in the form of a lexicalisation table like the one in (26), where the top row lists all the features and the other rows depict which morpheme lexicalises the relevant feature. When a feature is not present in a given form (Cond is absent in the past tense), this is depicted by putting black colour into the lexicalisation table.

(26) The lexicalisation table for X-marking and past forms of ‘feel’

	IPR	Asp	Mood	Cond	T	Past
feel, X-MARKING			fühl			te
feel, PAST			fühl			te

Let us now turn to the issue of how evacuation works. Evacuation movements are triggered in accordance with the lexicalisation algorithm provided in (27).

(27) The lexicalisation algorithm (De Clercq et al. 2025)

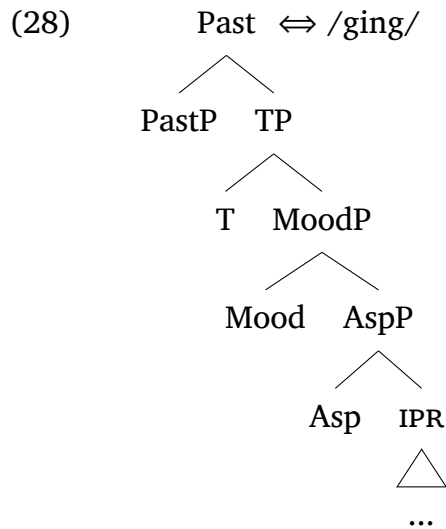
- a. Merge F and lexicalise
- b. If fail, evacuate the closest labelled non-remnant constituent and lexicalise
- c. If fail, evacuate the immediately dominating constituent and lexicalise (recursive)
- d. If fail, build a complex left branch
- e. If fail, backtrack to the previous cycle and try the next option for that cycle

The algorithm implements a cyclic procedure, where features are added (by merge) one by one, and after each feature F is introduced, the resulting phrase FP must be lexicalised, see (27a). This does not mean that the structure is immediately pronounced; the word ‘lexicalise’ in (27) means that a matching item must be found. If a matching item is found, the derivation ‘converges at the interface’ (it can be externalised), and another feature may be merged to the existing phrase marker (triggering lexicalisation again).

However, when matching fails, the FP created by merge F cannot be mapped onto PF (it cannot be externalised). In such cases, the derivation ‘crashes at the interface,’ and it is sent back to syntax. What syntax has to do is to perform evacuation movements, starting from the closest available constituent with the right properties, and retrying lexicalisation of FP after the constituent has evacuated it (see (27b)). If this still does not help, then evacuation targets larger and larger constituents (pied-piping), see (27c). The step (27d) turns out to be irrelevant for the present purpose, and the final step (backtracking) will be highlighted at the point where it becomes relevant.

Let me illustrate this on the derivation of the pair for ‘go,’ which is *ging(-st)* in the past tense and *ging-e(-st)* in the X-marking construction.⁵ Note that *ging* lacks the past tense *-te*, which leads me (as well as the child acquiring German) to posit the lexical entry in (28), where the features of the past tense are fully lexicalised by the root with no affix.

⁵Both of these are suppletive with respect to the present tense *gehe(-st)*, which is not discussed here for reasons of space. See Vanden Wyngaerd et al. (2021) for the discussion of suppletion and pointers, the device that is required to handle it.



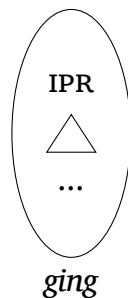
A simplified derivation of the past tense form – following the lexicalisation algorithm (27) – is shown in (29). It starts when the features corresponding to the Initiation, Process and Result are assembled, see (29a). Now a matching item must be found in the lexicon. The lexical item (28) contains, at its bottom, a constituent identical to (29a). Therefore, (29a) can be lexicalised by *ging*, see (29b). After successful lexicalisation, another feature is merged, see (29c). Lexicalisation is successful again, since a constituent identical to the AspP in (29c) is stored in the lexicon, namely as a subpart of the entry (28). Successful lexicalisation is depicted in (29d).

(29) The derivation of the past tense I

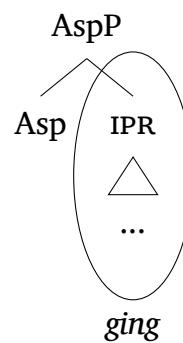
a. Merge F



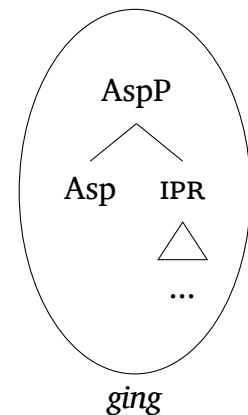
b. Lexicalise



c. Merge F



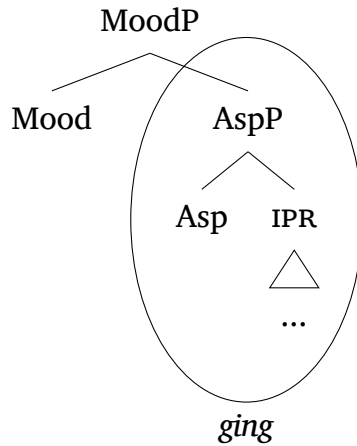
d. Lexicalise



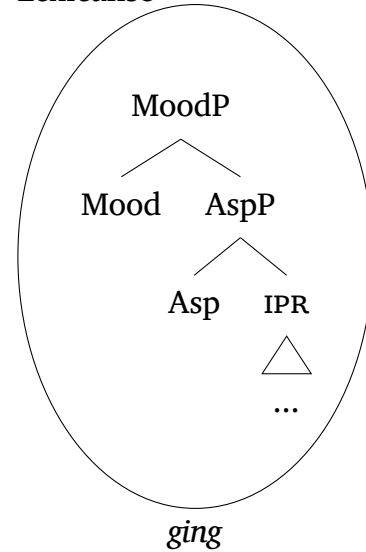
As the next step, Mood is added, as in (30a), and successfully lexicalised, as in (30b).

(30) The derivation of the past tense II

a. Merge F



b. Lexicalise

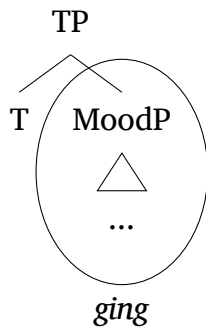


All the steps taken up to now are common for the derivation of the past tense and the conditional. At this point, the derivations diverge. While the derivation of the past tense merges tense right above MoodP (as derived in (30b)), the X-marking construction needs to add Cond first. Let us look at the derivations in turn.

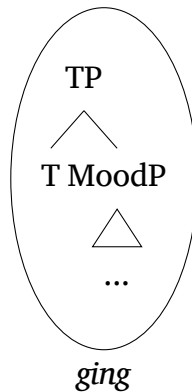
The derivation of the past tense is simple, merging the features T and Past, and always lexicalising with the root *ging*, see the steps in (31).

(31) The derivation of the past tense III

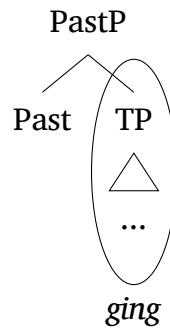
a. Merge F



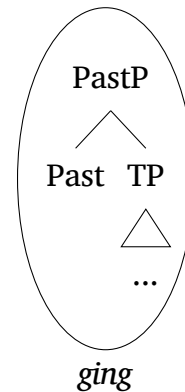
b. Lexicalise



c. Merge F

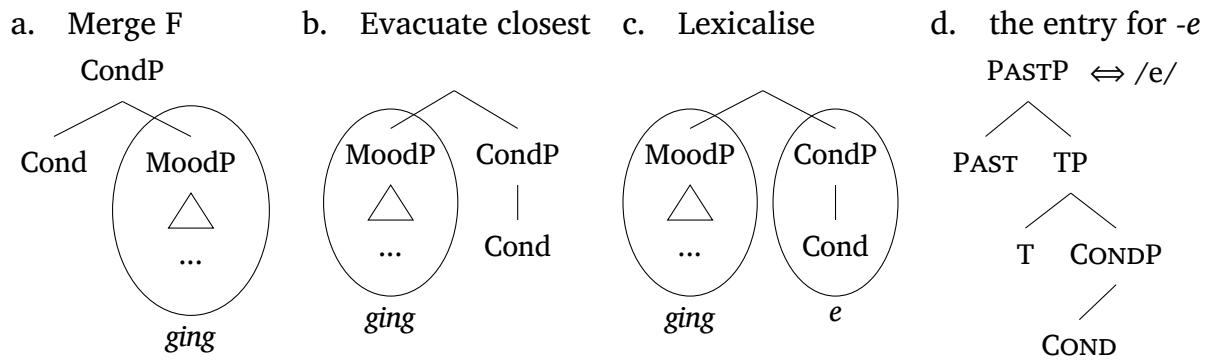


d. Lexicalise



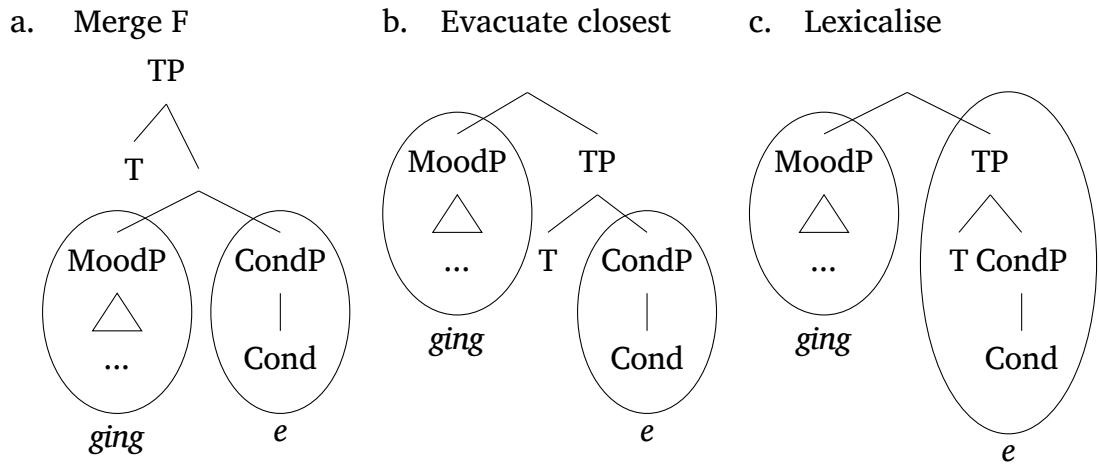
The situation is different for the conditional. When Cond is merged, as in (32a), there is no match, since the entry (28) does not contain Cond. Therefore, the closest labelled non-remnant constituent must be extracted out of CondP, see (32b). After the evacuation movement, the remnant CondP is matched by *-e*, as in (32c). The assumed entry for *-e* is in (32d).

(32) The derivation of the X-marking construction I



After Cond, we still have to add T and Past. When T is added, as in (33a), there is no match. Therefore, evacuate closest extracts MoodP out of TP, see (33b). After evacuation, the derivation is successfully lexicalised, see (33c).

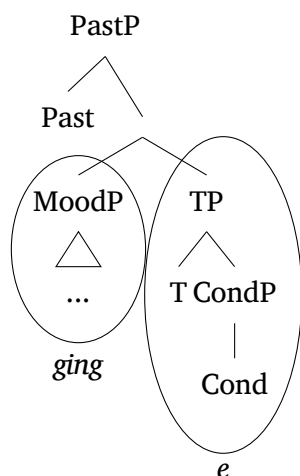
(33) The derivation of the X-marking construction II



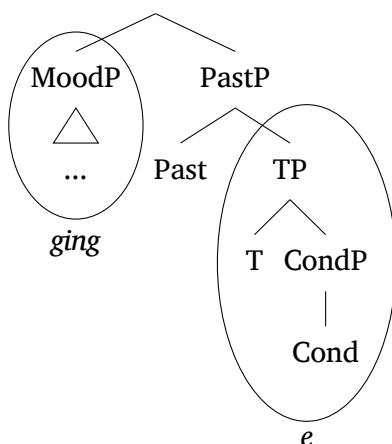
The same sequence of steps is repeated for Past, see (34). First, Past is merged (see (34a)). When no match is found, MoodP is evacuated out of PastP, see (34b). The resulting form *ging-e* is shown in (34c).

(34) The derivation of the X-marking construction III

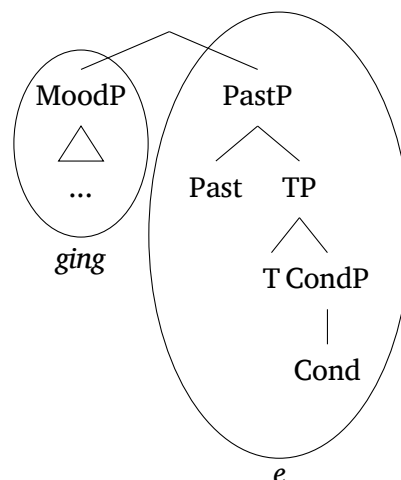
a. Merge F



b. Evacuate closest



c. Lexicalise



The resulting lexicalisation table is shown in (35), where the lexicalisation pattern of ‘feel’ is compared with ‘go.’ We can see that the root *fühl* always lexicalises the structure up to T, and the suffix *-te* starts at the T level. This is different for *ging*, which goes all the way up to Past in the past tense, but cannot lexicalise Cond (unlike *fühl*). As a result, the suffix *-e* must be used to lexicalise Cond, and it stretches all the way up to Past.

(35) The lexicalisation table for X-marking and past forms of ‘feel’ and ‘go’

	IPR	Asp	Mood	Cond	T	Past
feel, X-MARKING			fühl			te
feel, PAST			fühl			te
go, X-MARKING			ging		e	
go, PAST			ging			

As the last point in this section, it’s worth noting the fact that for each root in (35), the root lexicalises a different stretch of heads in the past tense and in the X-marking construction. For instance, *fühl* does not lexicalise Cond in the past tense, but it does so in the X-marking construction. The root *ging* lexicalises Tense in the past tense, but not in the X-marking construction. Thus, the fact that the roots are the same across the two environments is treated as syncretism; despite the fact that each root lexicalises a different set of features, its form is the same.

However, syncretism is never necessary, and we are thus not surprised that other roots, such as *brauchen* ‘need’ and *singen* ‘sing’ show differences across the two different forms, as in Table (36), repeated from (21).

(36) The forms of the past and X-marking (Majdić 2025)

	Present Ind.	Past	X-marked
feel	du fühl(-st)	du fühl-te(-st)	du fühl-te(-st)
need	du brauch(-st)	du brauch-te(-st)	du bräuch-te(-st)
go	du geh(-st)	du ging(-st)	du ging-e(-st)
sing	du sing(-st)	du sang(-st)	du säng-e(-st)

The idea is that the different root shapes for ‘need’ and ‘sing’ reflect the different number of features lexicalised by the root, as in (37). Otherwise the mechanics of the derivation is the same as for ‘go’ and ‘feel’, as is obvious from the comparison between the tables (35) and (37).

(37) The lexicalisation table for X-marking and past forms of ‘need’ and ‘sing’

	IPR	Asp	Mood	cond	T	Past
need, X-MARKING		brauch				te
need, PAST		bräuch				te
sing, X-MARKING		säng			e	
sing, PAST		sang				

3.3 Resolving the paradox

This section turns to the apparent mirror-violating order in Czech, with the relevant data in (38). The fact we are exploring is that when both morphemes follow the verb in Czech, the past-participle *-l* is always closer to the root than the conditional *by*, see (38b). The opposite order is impossible, see (38c).⁶

- (38) a. Přiše-l.
arrive-PAST.PART
‘He arrived.’
- b. Přiše-l **by**, pokud ...
arrive-PAST.PART COND if
‘He would arrive, if ...’
- c. *Přiše **by**-l, pokud ...
arrive COND-PAST.PART if
‘He would arrive, if ...’

The essence of the solution to the paradox is that the lexicalisation in the Czech conditional works almost exactly the same in the case of German ‘go’, repeated in (39).

(39) The lexicalisation table for X-marking and past forms of ‘go’

	IPR	Asp	Mood	cond	T	Past
go, X-MARKING		ging			e	
go, PAST		ging				

⁶Recall that the Czech *by* is a clitic, and it may also detach from the verb altogether. I shall discuss the clitic nature of *by* in Section 4.

The relevant property of the table is that the conditional *-e* apparently attaches on top of the ‘past tense.’ However, as the lexicalisation table makes it clear, this is not a true past tense form, but rather a form that only lexicalises a subset of the features of the past tense (up to Mood). In this analysis, it is (despite appearances) the *-e* that is lexicalising the Past feature in the form *ging-e*.

Now adapting this proposal for Czech, I am going to claim that Czech works the same and that the participle *nes-l* is ambiguous in the same way, just that it is bimorphemic. In this approach, *-l* lexicalises several features in the past tense, starting at Mood and going all the way to Past; see the bottom row in (40). However, its lexical entry is incapable of lexicalising Cond; this is why in the X-marked form, *-l* only lexicalises Mood and *by* lexicalises all the features from Cond up, just like *-e* does in German. As a result, *-l* is to the left of *by* in the surface string.

(40) The lexicalisation table for X-marking and past forms of Czech

	IPR	Asp	Mood	cond	T	Past
carry, X-MARKING	nes		1		by	
carry, PAST	nes		1			

The solution of the paradox is thus tantamount to the proposal that even though *-l* is a past tense marker in some of its uses, in the X-marking construction is not a true past tense marker, but a marker of some lower category, namely Mood in (40).

An independent piece of evidence for the idea that the marker *-l* is not always a past tense marker (but has also other, non-finite uses) comes from the fact that the morpheme is also used to form static participles from unaccusative verbs (see, e.g., Biskup 2016 for a discussion).

(41) Czech (based on Biskup 2016:9)

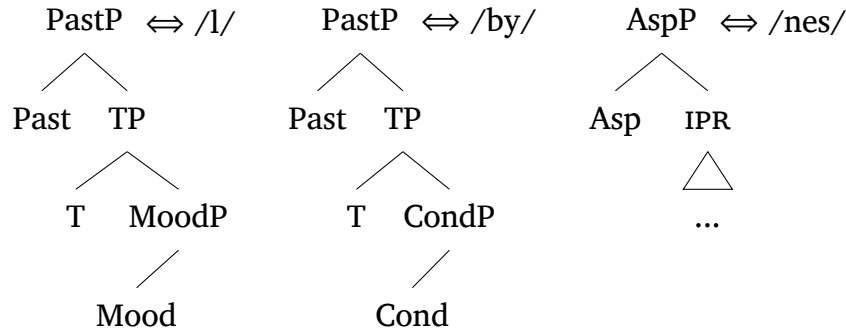
- a. o- mrz -l -ý prst
around- froze -PAST.PART -AGR finger
‘a frostbitten finger’
- b. roz- kvet -l -ý strom
apart- blossom PAST.PART -AGR tree
‘a tree in blossom’

Clearly, these are not past/finite forms, and they require, quite independently of the mirror-violating order, that the past participle *-l* receives an analysis where it also has other uses outside of the past tense. The table in (40) contains one such analysis where the affix is placed into Mood. In principle, this affix could go even lower and start, e.g., in the aspectual head; I leave this unresolved here. The important point is that the ambiguous nature of *-l* (sometimes expressing past, sometimes not) – which the analysis of the ordering paradox rests on – finds independent support in the existence of participles such as those in (41).

Let me now show how the forms in the lexicalisation table (40) are derived using the lexical entries in (42). What needs to be added is that the root (on the very right) is drawn as a triangle to save space, but I assume that it has a binary branching structure all the way down, which means that it requires no movement

for it to lexicalise the constituents it contains. On the other hand, *-l* has, at its bottom, a unary branching MoodP. Such a structure requires the complement of Mood to extract for *-l* to match MoodP (or any of the larger constituents). Since *-l* requires the root to extract to the left, it will necessarily follow it (and behave as a suffix). I assume the same for *by*: also here its complement will need to extract for matching to obtain, and so nothing else said, *by* will act as a ‘suffix’ (or more precisely, as a ‘post’ marker, leaving it open whether it is a clitic or an affix, but in any case following the base).

(42) Lexical items

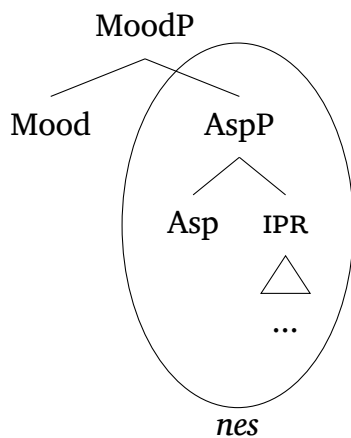


I shall now run the derivations under the assumption that both *-l* and *by* are ‘suffixes’ and return to the clitic status of *by* in Section 4. The main update in that section is going to be that even though *by* is a ‘post’ marker by default, it will also have access to more options than standard affixes (namely, it will be able to undergo clitic movement). With this in mind, let me turn to the derivations.

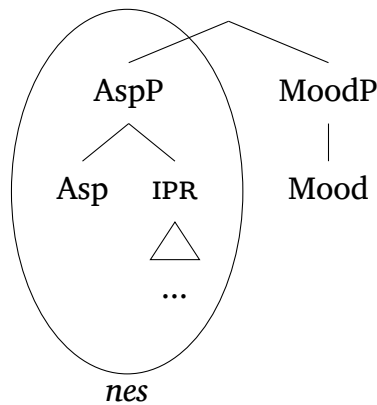
Both in the conditional and in the past tense, the derivation starts by assembling AspP, which is lexicalised by the root. (43a) shows a stage of the derivation where Mood has been introduced on top of the root. Assuming that the root *nes* ‘carry’ only lexicalises the AspP (see (42)), there is no match in the lexicon for (43a). Therefore, evacuate closest applies, and we get the tree in (43b). The remnant MoodP is lexicalised by *-l*, see (43c).

(43) The common part of the derivation

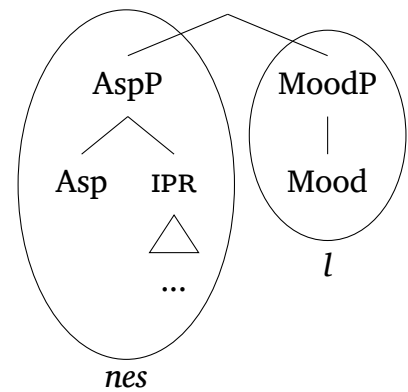
a. Merge F



b. Evacuate closest



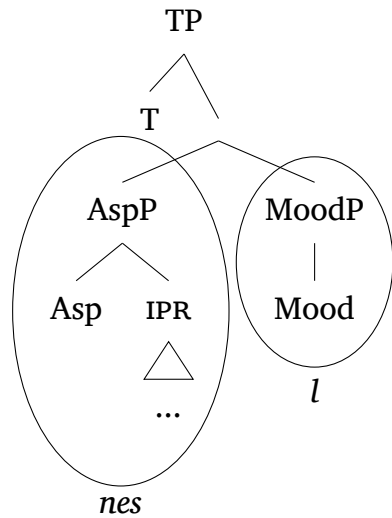
c. Lexicalise



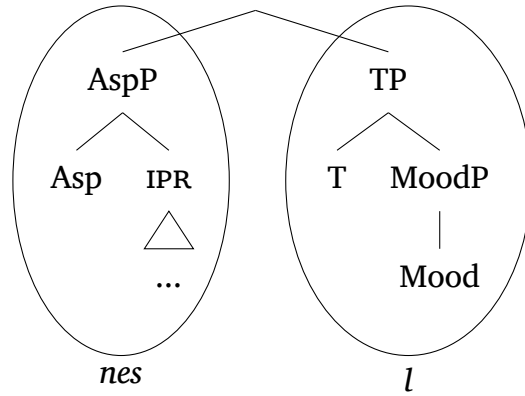
At this point the derivations diverge. In the past tense, T is merged directly on top of MoodP, see (44a). There is no match, so AspP (the closest labelled non-remnant node) is evacuated out of TP, see (44b). In (44b), we also see that *-l* matches the remnant TP (after AspP evacuates).

(44) The derivation of the past tense I

a. Merge F



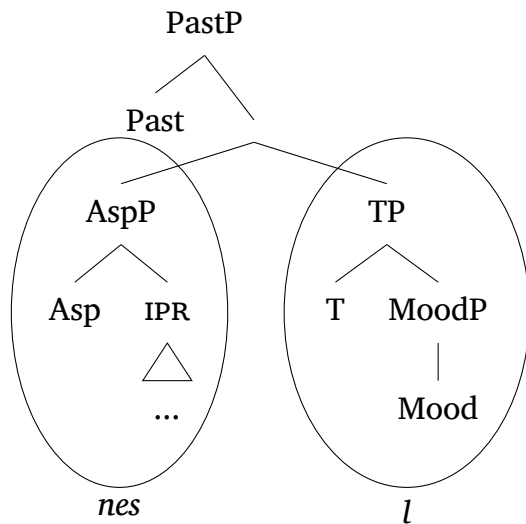
b. Evacuate closest + Lexicalise



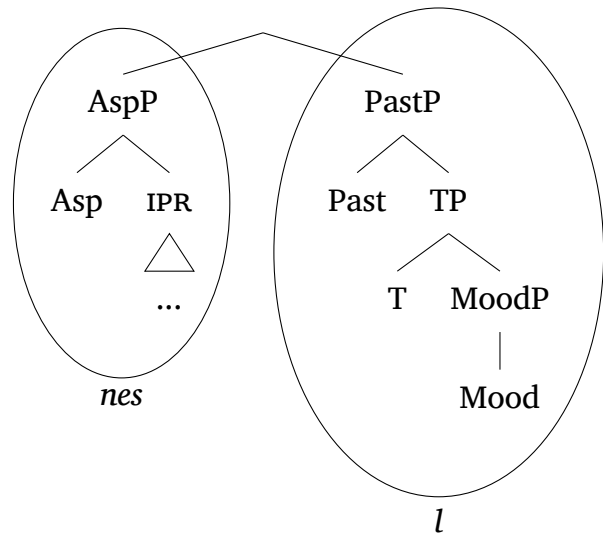
When Past is added, see (45a), there is no match. AspP (the closest labelled non-remnant node) is again evacuated out of PastP, see (45b). The past tense *-l* matches the remnant PastP and the past tense is correctly derived.

(45) The derivation of the past tense II

a. Merge F



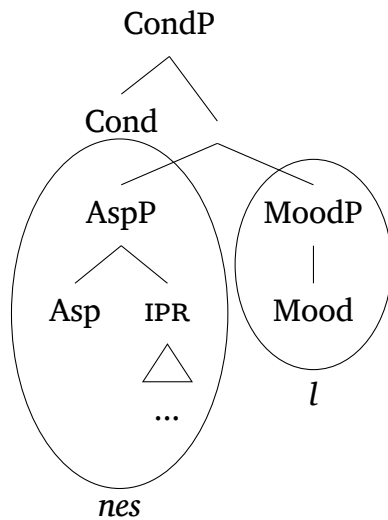
b. Evacuate closest + Lexicalise



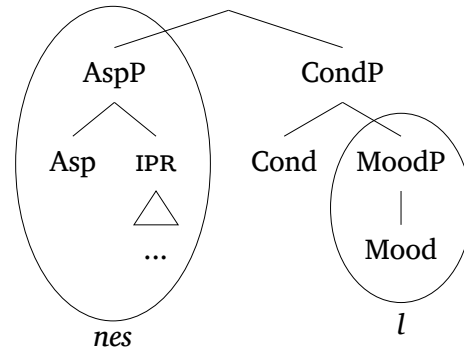
The X-marking construction needs to merge Cond on top of MoodP, see (46a). (Recall that the MoodP is the same one that we derived in (43)). This structure cannot be lexicalised (there is no match), so evacuate closest applies, yielding (46b). However, the remnant CondP finds no match either, recall the entries in (42). Therefore, the derivation returns to the stage (46a), and pied-pipes the next higher node above Asp, which yields the structure (46c). Here, there is a match for CondP, which is lexicalised by the matching item *by*.

(46) The derivation of the conditional I

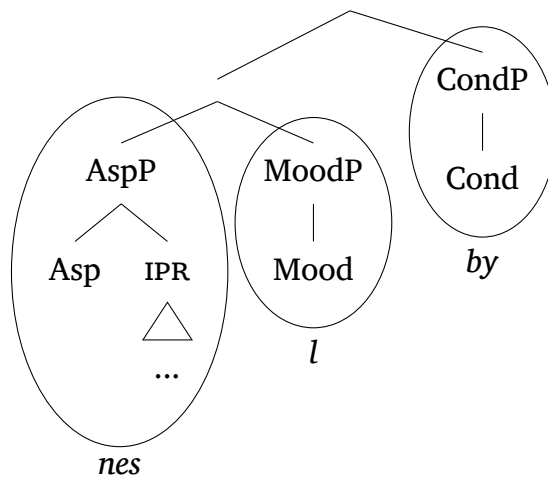
a. Merge F



b. Evacuate closest



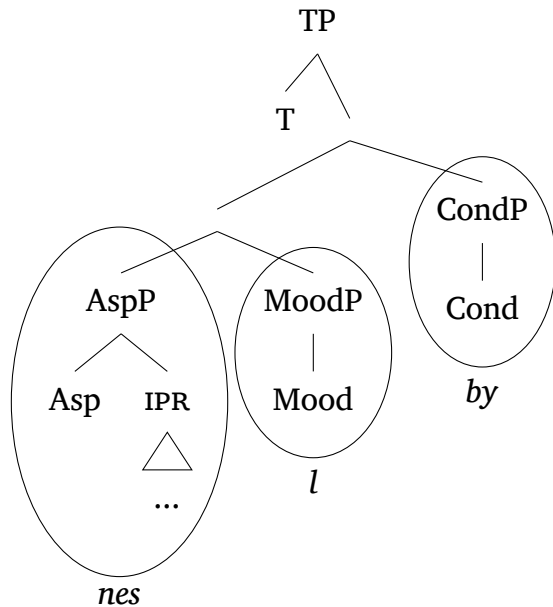
c. Pied-pipe + Lexicalise



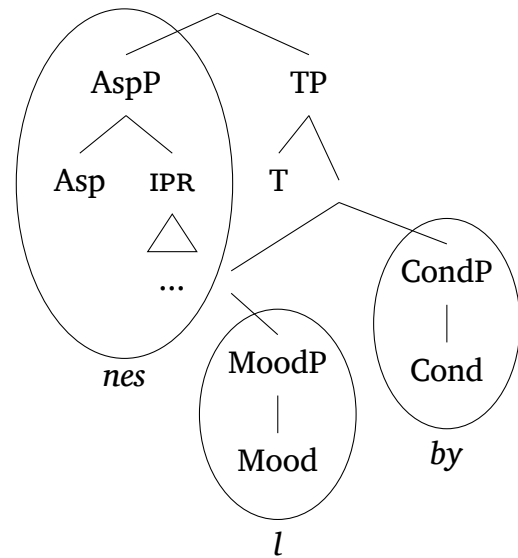
When T is added, there is no match (see (47a)). Therefore, the closest labelled non-remnant node is extracted, leading to (47b). But there is no match, so the derivation returns to (47a) and moves the next higher node (the pied-pipe step of the algorithm). The result is shown in (47c). Here, matching succeeds, and the whole TP is lexicalised by *by*, see (47d).

(47) The derivation of the conditional II

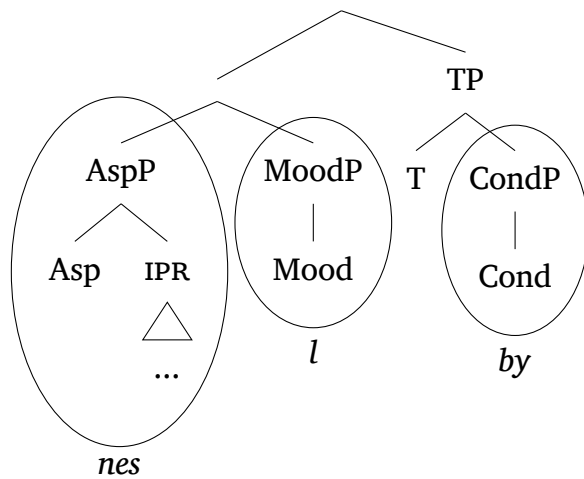
a. Merge F



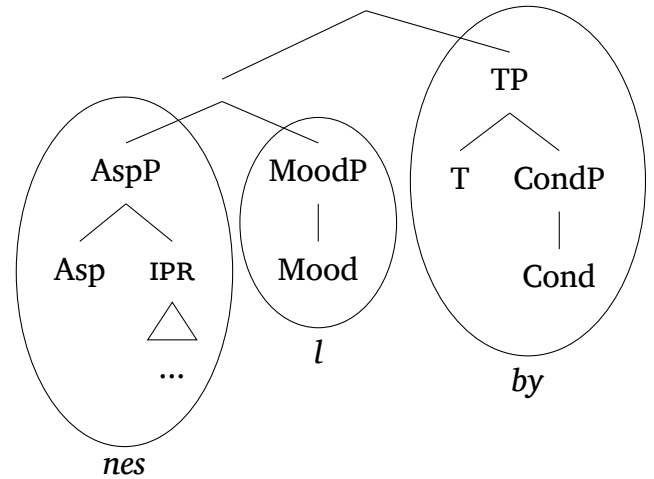
b. Evacuate closest



c. Pied-pipe

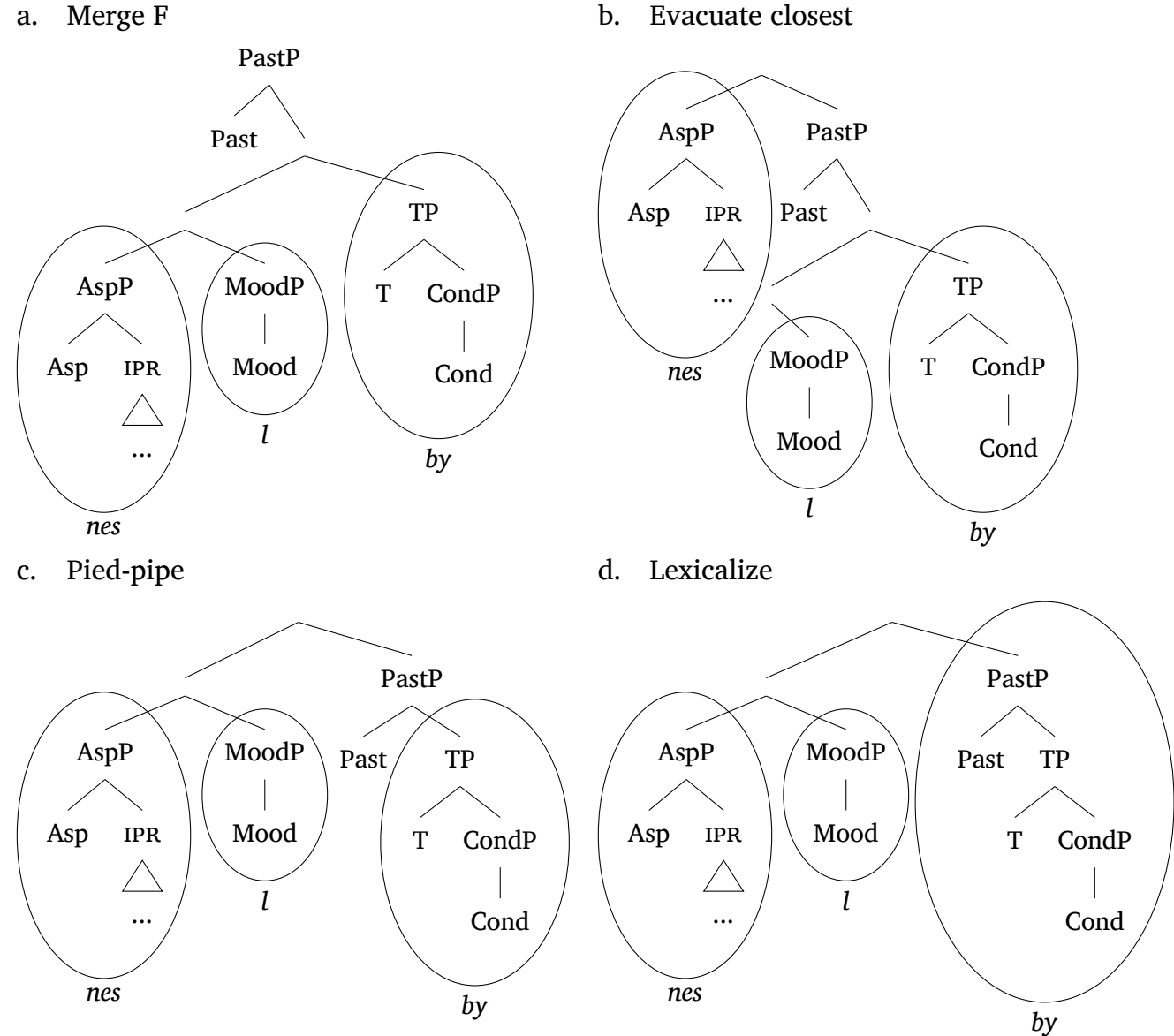


d. Lexicalize



To derive the X-marking construction, we still need to merge Past, see (48a). Since there is no match, we try evacuating the closest non-remnant constituent, see (48b). Since there is still no match, we need to go back to (48a) and pied-pipe the next higher node, yielding (48c). Here PastP is matched by *by*, see (48d).

(48) The derivation of the conditional III



As a result, we derive successfully the forms envisaged in the lexicalisation table (40), repeated in (49) for convenience.

(49) The lexicalisation table for X-marking and past forms of Czech

	IPR	Asp	Mood	cond	T	Past
carry, X-MARKING	nes		l			by
carry, PAST	nes		l			

What the lexicalisation table conveys is that *-l* is an ambiguous element. When it appears in the past tense, it is a portmanteau for Mood and Tense. This is

what gives it the aura of a past tense marker. However, when it appears in the X-marking construction, it is not a past tense marker: it lexicalises only Mood, and it is ordered with respect to *by* (that lexicalises Past in the X-marking construction) in a completely expected and predictable manner.

Before I leave this section, I would like to point out that the solution presented here is the same as the approach pursued in Sergienko (2025). Sergienko looks at the past and perfect forms in Ishkashimi, and observes that the perfect form is apparently based on the past tense. This is shown in Table (50), where the past tense *-d* is followed by the perfect *-ûk*; NPST refers to a ‘non-past’ form.

(50) An Ishkashimi paradigm of *wenûk* ‘to see’ (Sergienko 2025:206)

NPST	PST	PRF
wen	wen-d	wen-d-ûk

The order V-Past-PERF in Ishkashimi (*wen-d-ûk*) is the mirror image of the common order V-PERF-Past (Cinque 2014), as found, among others, in Turkish, see (51).

(51) Turkish (Kornfilt 2013:350)

- a. ben lokanta-ya var -dığ -ım -da,
I restaurant-DAT arrive -NOMIN. -1SG -LOC
- b. Hasan yemeğ -in -i bitir -miş -ti.
Hasan meal 3SG ACC eat PERF PAST
‘When I arrived at the restaurant, Hasan had finished his meal.’
- c. Hasan yemeğ -in -i bitir -miş ol -acak.
Hasan meal 3SG ACC eat PERF be FUT
‘When I arrive at the restaurant, Hasan will have finished his meal.’

Just like in Czech, there is a contrast in the order of suffixes suggesting an apparent mirror-principle violation. Sergienko’s analysis of the mirror-violating form *wen-d-ûk* in Ishkashimi (compared to the past tense *wen-d*) is shown in the (52).

(52) The lexicalisation table for X-marking and past forms of Czech

	IPR	Asp	Perf	T	Past
see, PAST	wen	d			
see, PERF	wen	d	ûk		

Looking at the table, we can see that Sergienko’s idea is that in the past tense, *-d* actually lexicalises all the projections from Asp to Past, see the first row in (52). However, when the derivation contains perf, *-d* only lexicalises Asp, and the features perf and T are lexicalised by *-ûk*. This approach thus generalises to multiple instances of mirror-violating orders: a particular tense marker (*-l* in Czech, *-d* in Ishkashimi) is in reality a portmanteau, and in some forms, it lexicalises only the lower category (e.g., just Asp or just Mood). In such cases, the marker stops behaving like a tense marker and gets ordered as a Mood/Asp marker, giving rise to apparent mirror-violations. Upon closer inspection, however, these violations are only an illusion.

4 The clitic nature of *by*: a tentative proposal

In the previous section, I showed how the sequence VERB > PAST PARTICIPLE > COND, seen in (53a), can be generated.

- (53) a. *Přiše-l by (dnes), pokud ...*
 come-PAST.PART COND today if
 ‘He would come today, if...’
 b. *Petr by (dnes) přiše-l, pokud ...*
 Petr COND today come-PAST.PART if ...
 ‘Petr would come today if ...’

What remains to be explained is how this proposal generates examples where the verb is not the first constituent, as in (53b). In this example, there is an overt subject, and the clitic assumes a second position after the subject. The clitic can be separated from the verb by an adverbial – ‘today’ in (53b). (The clitic cannot be separated from the participle in (53a), as that would place it third.)

The tentative proposal I suggest here is that the order in (53a) is basic, and the order in (53b) is derived by clitic movement. In this approach, *by* is born as a marker that follows the verb (a post-marker), exactly as proposed in Section 3.3. However, it is a special element (‘a second position clitic’), and it is only seen in its base position as long as the second position requirement is satisfied, i.e., when the verb is first. When the second position requirement is not met, the clitic has to move to the second position, as shown in (54). Here, the strike-through indicates the ‘base’ position of *by* within a constituent that is derived by the lexicalisation movements as described in the previous section. Since in this configuration (when the verb is not first), the second position requirement is not satisfied, *by* leaves the verb and moves to the second position.

- (54) *Petr by dnes [přiše-l ~~by~~], pokud ...*
 Petr COND today come-PAST.PART ~~COND~~ if ...

This approach is compatible with approaches such as Bošković (2001), Marušič (2008) or Dotlačil (2008), where the surface position of clitics is not always the same position. For instance, one could posit that the clitic always moves out of the base position, but when there is no XP preceding, it is the lower copy that is pronounced for prosodic reasons, as in (55).⁷

- (55) *by [Přiše-l by], pokud ...*
 COND come-PAST.PART COND if

Leaving this issue aside, the current section considers some beneficial consequences of this proposal, even though a full discussion is beyond the scope of the chapter. In Section 4.1, I consider the ordering of *by* and negation, suggesting that the current proposal allows us to maintain a single position for Neg

⁷In this approach, the postverbal *by* in examples such as (55) have a special status. Since the clitic is in the second position without any need for movement, it basically acts as a suffix-like element. This seems to be supported by the results reached in Polómská & Ziková (2026), who show that clitics in Czech interact prosodically differently with their host depending on what the host is. When it is the verb, the clitics are more closely integrated with the host than following any other host.

across Slavic. In Section 4.2, I consider clitic doubling as evidence for movement. Section 4.3 concludes.

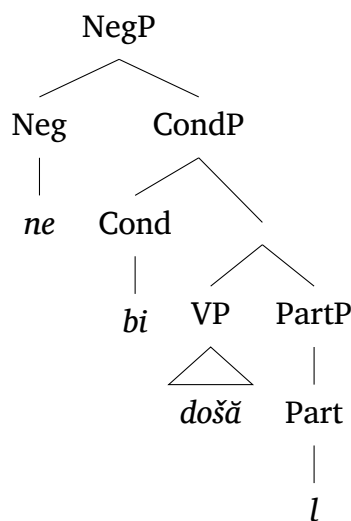
4.1 Verbal clitics and negation

One of the reasons why I propose that the clitic *by* is underlyingly a post-marker is related to the difference in the placement of negation in Czech and some of the South Slavic languages, including Bulgarian. To begin with, it is relevant to note that in all Slavic varieties, negation precedes the verb. However, variation concerns the placement of the conditional *by* (Bulgarian *bi*). Specifically, in Bulgarian, the conditional *bi* intervenes between the negation and the verb, see (56a). In Czech, the negation always directly precedes the participle, see (56b) (it can never be separated from the participle by any auxiliary – including the conditional *by*).

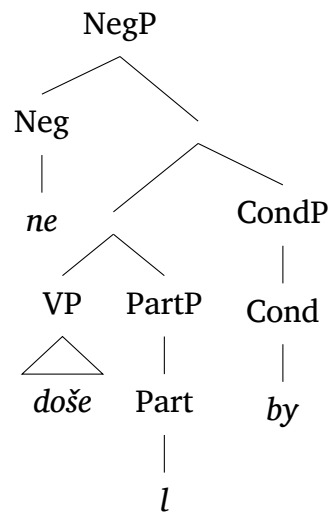
- (56) Bulgarian (a) vs Czech (b)
- a. **ne bi** došā-l, ...
NEG COND come-PAST.PART
 - b. **ne-** doše-l **by**,
NEG come-PAST.PART COND
- both: ‘He would not come, ...’

The simplest analysis is one where negation has the same scope in the two languages, i.e., one where Neg always scopes over the conditional *bi/by*. The difference can then be reduced to the proposal that *bi/by* is a pre-marker in Bulgarian, but a post-marker in Czech. The two structures are provided in a simplified format in (57) and (58), respectively. The Czech structure in (58) is a simplified version of what was argued in Section 3.3, i.e., we have the verb moving across the participle marker, and then the whole constituent moves across the conditional marker. In Bulgarian, shown in (57), the movement of the constituent containing the verb and the participle is absent, hence the difference.⁸

(57) Bulgarian



(58) Czech



⁸In Nanosyntax, pre-markers are constructed in an independent workspace and joined as a ‘complex left branch’ to the main spine.

The orders in (57) and (58) are as found in (56). Now recall that when the verb cluster is not sentence initial, I assume that in Czech, the clitic *by* is found in a dislocated position, moving out of the verb cluster, on analogy to pronominal clitics (see Toman 1999, Lenertová 2004, Tabachnick 2024). This is the case in a sentence like (59a), with the analysis depicted in (59b).

(59) Czech

- a. Petr **by** (dnes) **ne-** doše-l, kdyby ...
 Petr COND today NEG come-PAST.PART if ...
 b. Petr **by** (dnes) [**ne-** doše-l **by**], kdyby ...
 Petr COND today NEG come-PAST.PART COND if ...

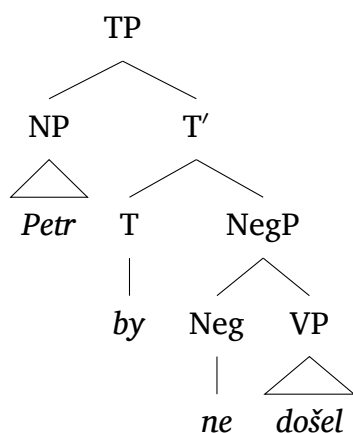
As a result of the movement, the conditional clitic crosses other elements of the verb cluster, including the negation (and potentially adverbs, recall (53b)). This creates a contrast with Bulgarian, which has adverbial clitics. Under my analysis, this means that Bulgarian clitics do not move and remain within the verb cluster. Since they do not move, they always follow the negation, as in (60).

- (60) Toj **ne bi** doša-l,
 He NEG COND come-PAST.PART ... (Bulgarian)

The proposed analysis allows us to unify the underlying position of the negation relative to elements such as the conditional *by/bi* in the two languages. On my account, the negation always scopes over the conditional, but in Czech, clitics may move to the second position across it. Since Bulgarian clitics are adverbial (not second position), this movement is absent and *bi* always follows *ne*.

Without postulating the movement of verbal clitics in Czech, we would need to conclude that the conditional *by* is higher than Neg in the functional sequence. This account has been pursued in, e.g., Veselovská & Karlík (2004:184), who assume that verbal clitics do not move. In such an account, the facts are handled by proposing that Czech verbal clitics are generated higher than the Neg head, as in (61); (based on Veselovská and Karlík's account).

- (61) A structure with Cond above Neg (not adopted here)



This account works for Czech, but requires different base positions for negation across different languages: below the conditional in Czech, and above it in Bulgarian. On the other hand, if verbal clitic moves to the second position, this

allows for a unified hierarchy across Slavic (with Neg higher than Cond).

To summarise, this section argued that analysing *by* as a post-marker in Czech explains the contrast in the ordering between Czech and Bulgarian in sentences where the verb cluster is initial. In such sentences, the Czech *by* always follows the participle, while in Bulgarian, it precedes the participle. When the verb cluster is not first in the sentence, I postulate the existence of clitic movement in Czech, whereby *by* moves to the second position. This movement is absent in languages such as Bulgarian, which have adverbial clitics. This allows for a unified position of negation in the functional sequence.

4.2 Doubling

The second argument in favour of treating second position verbal clitics in Czech as undergoing movement comes from auxiliary doubling. To provide some background for the argument, consider first the fact that in some Slavic languages, second position clitics are turning into suffixes, losing their original second position status and remaining attached to the participle. This is, for instance, the case in Polish, as the following examples illustrate (see, e.g., Embick 1995, for a discussion).

(62) Polish (Migdalski 2010:350)

- a. Rano wyjechali -śmy z Wrocławia.
morning leave.PAST.PART -1PL from Wrocław
'We left Wrocław in the morning.'
- b. Rano wyjechali -by -śmy z Wrocławia.
morning leave.PAST.PART COND -1PL from Wrocław
'We left Wrocław in the morning.'

In Czech, such sentences require the clitics to move to second position, as shown in (63).

(63) Common Czech

- a. Ráno jsme [vyjeli jsme] z Bratislavi.
morning 1PL leave.PAST.PART 1PL from Wrocław
'We left Wrocław in the morning.'
- b. Ráno by jsme [vyjeli by jsme] z Bratislavi.
morning COND 1PL leave.PAST.PART COND 1PL from Wrocław
'We would leave Wrocław in the morning.'

However, in spoken Czech, there is one auxiliary that seems to be losing its second position status, namely the second-person past-tense auxiliary *-s*. Preininger (2017:36) looks at the uses of this marker in the corpus of spoken Czech and concludes that when this marker is not in the second position, it is overwhelmingly suffixed to the participle. Its second position use is illustrated in (64a), and its verb-adjacent use is illustrated in (64b).

(64) Czech variation in positioning the 2SG auxiliary

- a. Ten film s už vidě -l ?
the movie 2SG already see -PAST.PART

- b. Ten film už vidě -l -s?
 the movie already see -PAST.PART -2SG
 Both: 'Have you already seen that movie?

These constructions are interesting in that they show that when the marker loses its second position status, it occurs *suffixed* to the participle (rather than preceding it). This can be understood as a simple loss of the second position requirement: once the marker loses its second position status, it surfaces in its presumed base position.

With this background in place, consider the fact that Preininger (2017:42) also reports 54 examples in the spoken corpus where the auxiliary is doubled, so that one occurrence of it is in the second position, and the second occurrence on the participle, as in (66).⁹

- (66) Ten film s už vidě -l s?
 the movie 2SG already see -PAST.PART -2SG
 'Have you already seen that movie?

Obviously, the existence of such doubling highlights the need to consider two positions where the marker can appear, and these are precisely the two positions that are proposed here: the postverbal one and the second position. For some general ideas supporting the view that doubling and movement are related, see Barbiers et al. (2010), Barbiers (2014).

4.3 Summary

In this section, I tentatively suggested that verbal clitics in Czech originate adjacent to the verb, akin to clitics in South Slavic, which tend to be adverbial. However, unlike the Bulgarian conditional, these clitics are post-verbal, which can be captured by assuming that the participle simply moves across them inside the verbal cluster. The underlying post-verbal nature of these clitics is suggested by the relative positioning of negation, the verb and the clitics themselves. I also argued that when the relevant clitics lose their second position status, they appear after the verb in Czech (recall the 2ND person -s). The underlying position after the verb and the second position are arguably linked by movement, leading to examples of clitic doubling attested in spoken Czech.

5 Deriving the singular paradigm

This section builds on the previous results and extends the account to person/number marking. The full paradigm is repeated in (67) (recall (2)), though for reasons of space, this section only provides an analysis of the singular. For simplicity, I shall also ignore the fact that the participle agrees in gender.

⁹Preininger provides the following example from the corpus ORAL 2013:

- (65) ty s šla s k maturitě takto rozčuchaná ?
 you 2SG went 2SG to exam so disheveled

(67) Standard Czech

SINGULAR		PLURAL	
PAST	CONDITIONAL	PAST	CONDITIONAL
1 nes-l js-em	nes-l by-ch	nes-l-i js-me	nes-l-i by-chom
2 nes-l js-i	nes-l by-s	nes-l-i js-te	nes-l-i by-ste
3 nes-l	nes-l by	nes-l-i	nes-l-i by

In my analysis, I shall treat all the markers as suffixes, i.e., generating linear orders of the kind in (68).

- (68) a. verb > past participle > auxiliary > ϕ (1ST)
b. PAST nes -l js -em
c. CONDITIONAL nes -l by -ch

As highlighted in the previous section, since the auxiliaries are second position clitics, this order is only found when the verb is the first element in the sentence. When this is not the case, the clitics move to the second position.¹⁰

The account I suggest aims to capture the fact that the past tense and the conditional have different ϕ endings. However, we also need to keep in mind that in spoken Czech, the two paradigms undergo levelling, and the ϕ markers become identical across the two paradigms.

In (69), I show the lexicalisation table for the singular of the past tense and the conditional respectively. Compared to the account given in Section 3, the table includes an explicit proposal how person and number are lexicalised. Importantly, the analysis of the third person of the past tense maintains the idea that *-l* is the lexicalisation of Past, which, however, only lexicalises Mood in the conditional (due to the presence of Cond).

(69) The lexicalisation table for the singular

SPI	Asp	Mood	Cond	T	Past	SG	π	PART	SPKR
nes		1			1				
nes		1			js			i	
nes		1			js			em	
nes		1	by						
nes		1	by			s			
nes		1	by			ch			

The new thing in the table is what happens beyond the third person. There are two noteworthy features. The conditional *by* and the past participle *-l* lexicalise all the features of the 3RD person. In 1ST and 2ND person, they have to ‘shrink’ and lexicalise less than their full potential. I will show how this happens when we go through the derivations. In these persons, where *-l* ‘shrinks’ down to Mood, *js* and *by* emerge as auxiliary roots. Importantly, they each reach a different height in the lexicalisation table. While *js* lexicalises Tense and SG, *by* only goes up to

¹⁰Whether each of the clitics moves on its own – or whether they move inside a single constituent that the participle moves out of – is an issue that I shall not address here.

Past (and does not lexicalise SG). This leads to the fact that there are different sets of features left for the lexicalisation by the ending, leading to two different sets of ϕ markers for the two different auxiliary roots.

Arguably the most technical part of the analysis is associated to the participial marker *-l*. What I propose in (69) is that this marker lexicalises the past tense plus ϕ in the 3RD singular, only to retreat under Mood in the 1ST and 2ND persons. In these persons, the participial marker has the same status as in the conditional. Before we embark on the technical details, I would like to provide some non-technical motivation for this analysis.

The motivation consists in a kind of an ‘in between’ status of the Czech past tense, where the 3RD person has a different status than the 1ST and 2ND person. This contrast emerges when we compare present-day Standard Czech to Old Czech on the one hand and Polish on the other hand. As can be seen in Table (70), the Old Czech forms show that the past tense developed from a fully periphrastic construction, with the auxiliary root *j(e)s*, and a full set of ϕ endings *-em*, *-i*, *-t*. In present-day Literary Czech, the first and second person still pattern with Old Czech, but the 3RD person auxiliary was lost.

(70) Past tense in three varieties of Slavic¹¹

	Old Czech	Standard Czech	Polish
1	nes-l js-em	nes-l js-em	nios-ł-em
2	nes-l js-i	nes-l js-i	nios-ł-eś
3	nes-l jes-t	nes-l	nios-ł

In the last column, we can see that Polish developed a synthetic past tense paradigm across all persons. In these forms, the root of the auxiliary has been eliminated in all persons, and only the person/number suffixes remained. The Czech 3RD person patterns with Polish in this respect.

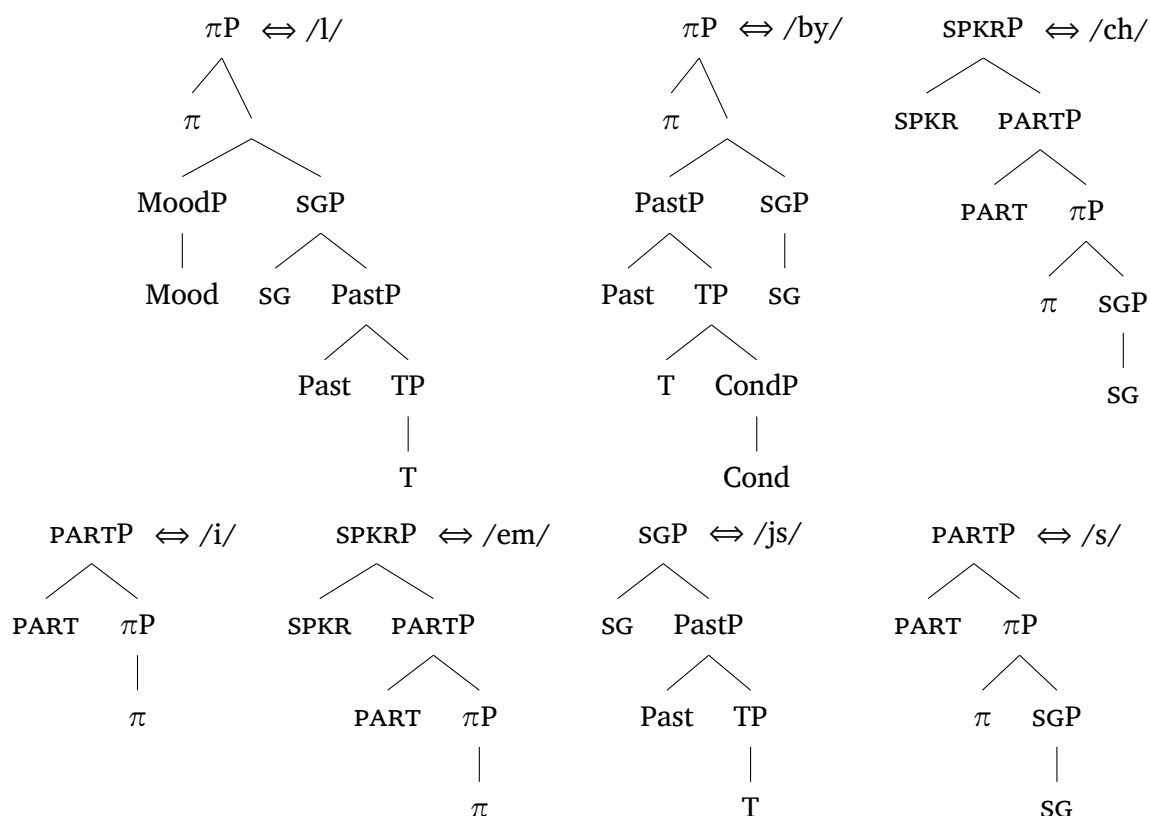
Czech thus has the most complicated paradigm, since it combines features of both of the systems, i.e., it contains both the original periphrastic system (in 1ST and 2ND person) and the diachronically newer past tense (in the 3RD person) that lacks the auxiliary root *j(e)s*, as found in Polish.

The Polish system would be much simpler to analyse because the emergence of the copular root *js* in Table (69) is non-trivial. To see why, recall that in Section 3.3 we saw that lexical items may ‘shrink’ and lexicalise just a part of their specification. In Table (69), it would be expected that in the 1ST and 2ND persons of the past tense, *-l* will be able to fill the inflectional space up to the person endings, leaving no room for the copula. This expectation is born out in the Polish paradigm, which has completely eliminated the copular root and left only the person/number endings. So the expectation is not wrong in the strict sense: we just need to be clever in setting up an analysis that can also account for Czech.

The lexical items needed for the analysis of the Czech paradigm are provided in (71).

¹¹Old Czech from Lamprecht et al. (1986:244), Polish from Swan (2002:252). I assume that the Old Czech auxiliary is *j(e)s*, where *-e* only surfaces when the auxiliary is not followed by a vowel.

(71) Lexical items



We can see that except for *-l* and *by*, all the entries are relatively straightforward, i.e., specified for all their features in the usual hierarchy as discussed in Section 3.1. The only exceptions are *-l* and *by*, which include the so-called complex left branch: see the MoodP between π and SG in the case of *-l*, and PastP in the same place in the case of *by*. These complex left branches – MoodP and PastP respectively – correspond to the main use of these markers: as a participial marker in the case of *-l* and an X marker in the case of *by*. The surrounding features (mostly ϕ features) are those that these markers acquired after the disappearance of ϕ morphology. Thus, the past participle *-l* has been originally only specified for Mood, but when the auxiliary *js* disappeared in the 3RD person, *-l* overtook its function: note that the sister of MoodP is identical to the root of the auxiliary, and the π of the 3RD person is found on top of the entry for *-l*. The same holds for *by*, which, in addition to its conditional role (as given on the left branch), lexicalises π and SG, as these are not overtly marked in 3RD SG form.

The ϕ suffixes are of two types. One type has SG as its lowest feature, these combine with *by*. Then there are pure person markers that have π as their lowest feature (*-em*, *-i*), these combine with *js*.

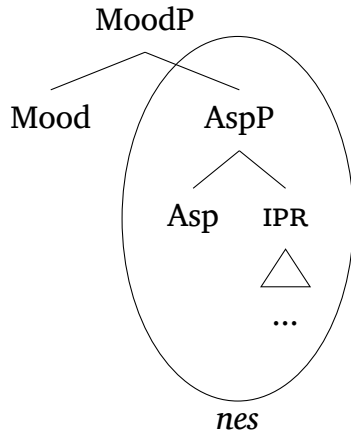
5.1 The derivation of the past tense

This section shows how the lexicalisation of the past tense in table (69) is derived using the entries in (71). The derivation starts by assembling AspP, which is lexicalised by the root. (72a) shows a stage of the derivation where Mood has been introduced on top of the root. Assuming that the root only lexicalises the AspP, there is no match in the lexicon for (72a). Therefore, evacuate closest

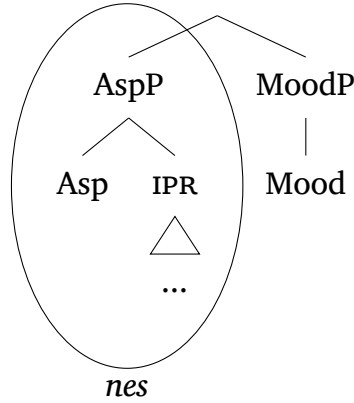
applies, and we get the tree in (72b). The remnant MoodP is lexicalised by *-l*, see (72c).

(72) The derivation of the past tense I

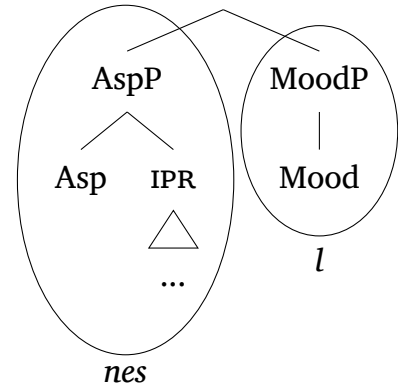
a. Merge F



b. Evacuate closest



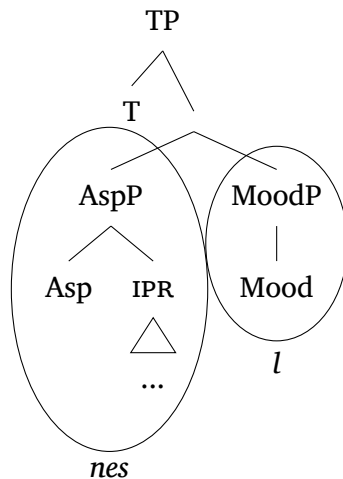
c. Lexicalise



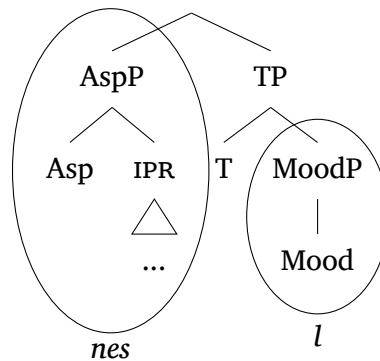
In the next step, we introduce the T head, see (73a). Since there is no match, evacuate closest targets AspP, see (73b). However, there is no match for the TP in (73b), so the movement is undone and we go back to (73a). Since evacuating AspP did not help, we pied-pipe the next higher phrase out of TP, yielding (73c). Here, lexicalisation succeeds and *js* is inserted as the lexicalisation of TP. Note that the marker *-l* is also a match for the TP in (73c), but it has more superfluous features, so it loses in competition to *js*.¹²

(73) The derivation of the past tense II

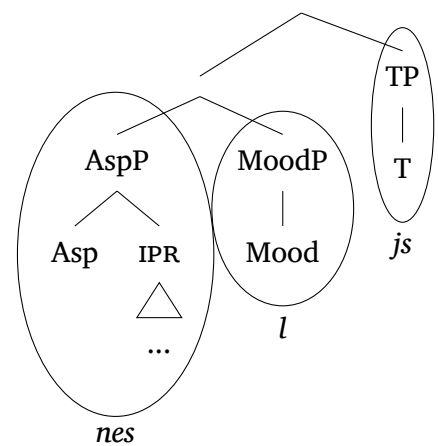
a. Merge F



b. Evacuate closest



c. Pied-pipe + Lexicalise

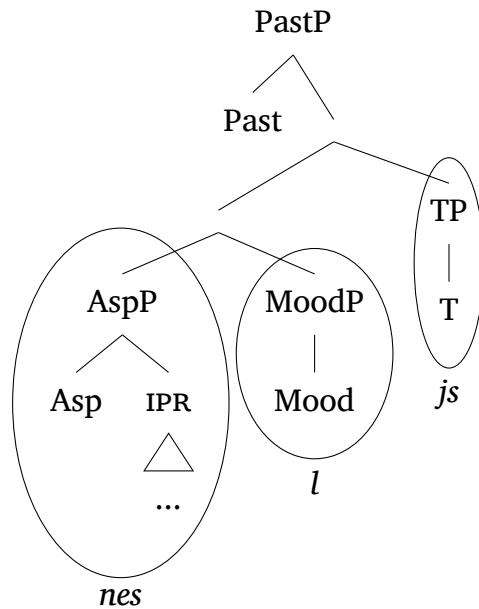


¹²It is still possible that *js-* will be replaced at a later stage. The lexicalisation procedure only accesses the lexicon at this stage to determine if there is a match, but nothing is sent to PF as yet.

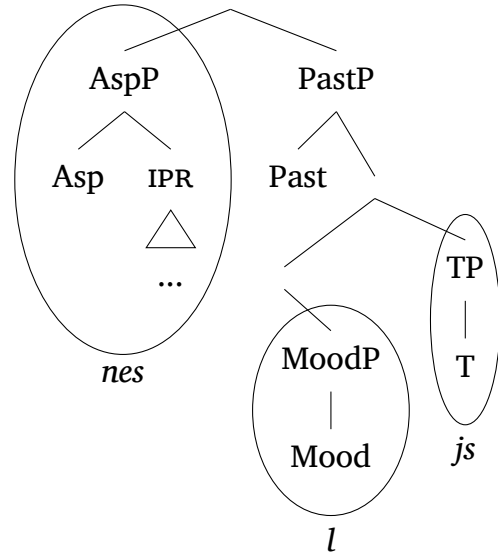
When Past is added to (73c), we get (74a). Evacuate closest produces (74b), but there is no match for the PastP in (74b).

(74) The derivation of the past tense III

a. Merge F



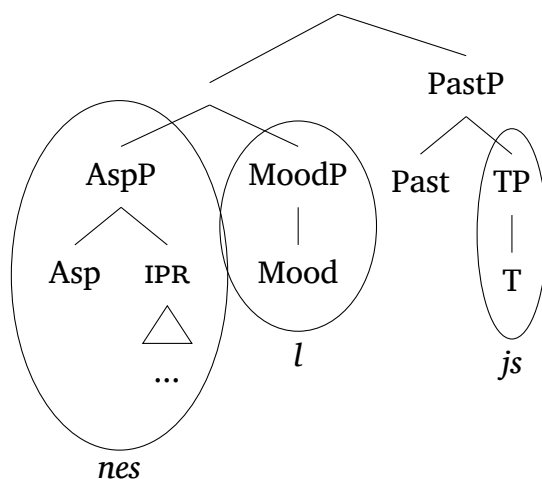
b. Evacuate closest



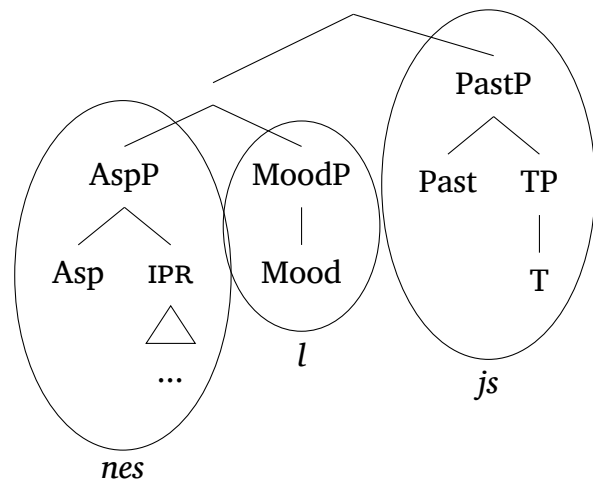
Therefore, the derivation goes back to (74a) and evacuates the next larger constituent, yielding (75a). Here lexicalisation succeeds, see (75b). (Again, *-l* could also lexicalise PastP, but it loses in competition.)

(75) The derivation of the past tense IV

a. Pied-pipe



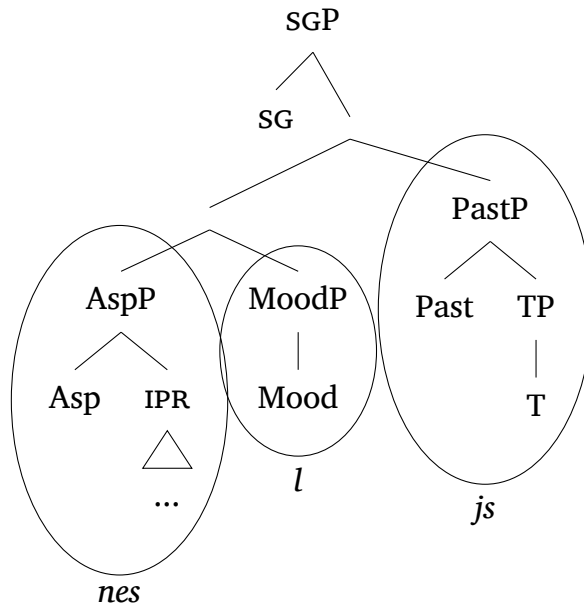
b. Lexicalise



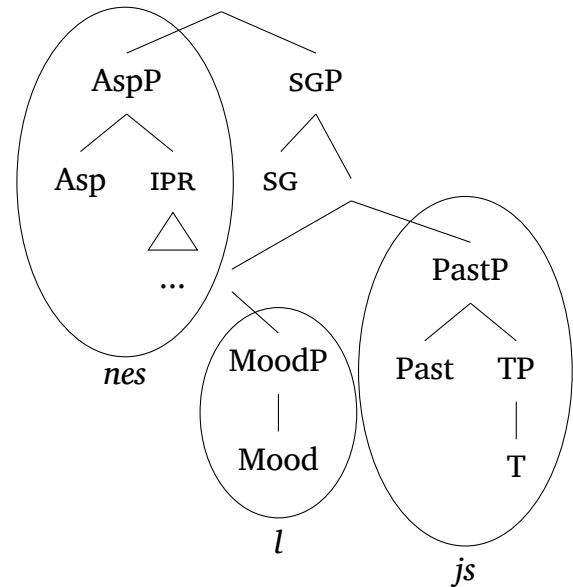
When SG is merged, see (76a), there is no match. Evacuate closest yields (76b), but we still have no match for SGP.

(76) The derivation of the past tense V

a. Merge F



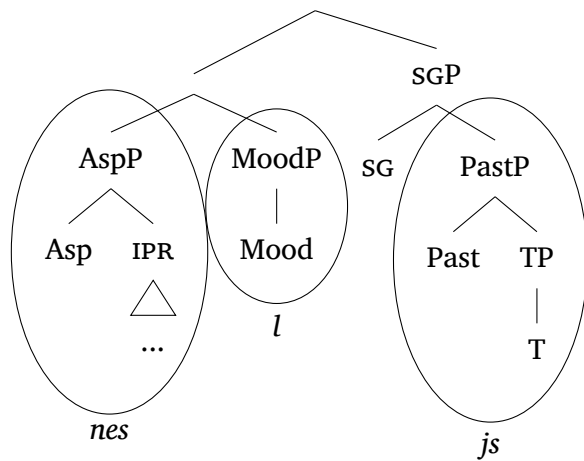
b. Evacuate closest



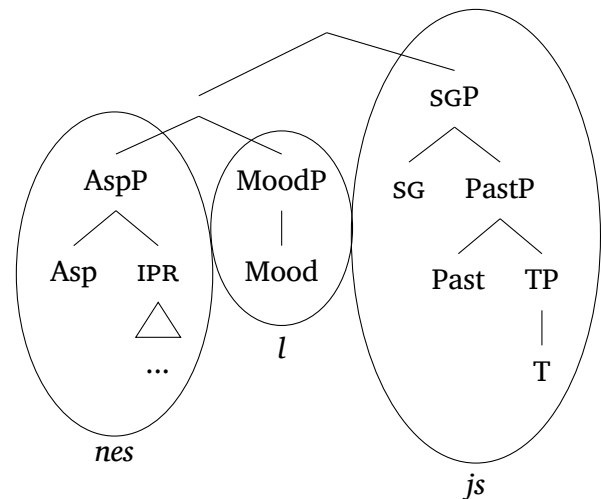
Therefore, the derivation goes back to (76a) and evacuates the next higher node above AspP, yielding (77a). Here SGP is lexicalised by *js*-, see (77b).

(77) The derivation of the past tense VI

a. Merge F



b. Pied-pipe

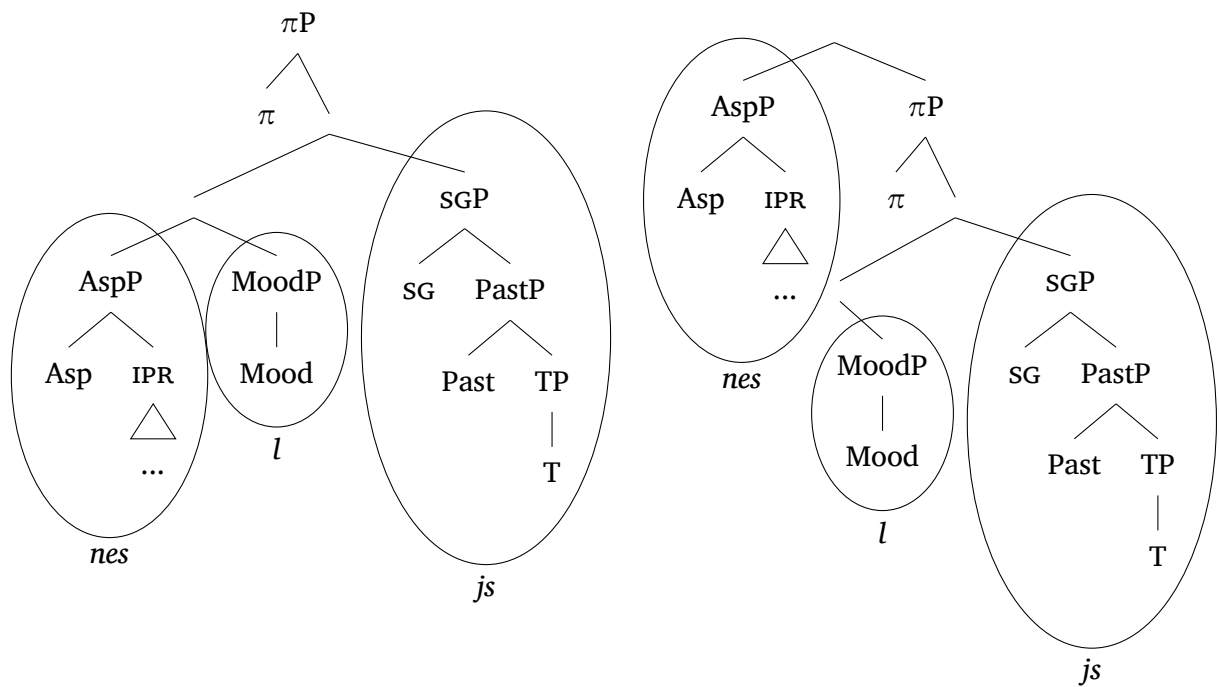


When π is merged on top of (77b), we get (78a). Evacuate closest produces (78b).

(78) The derivation of the past tense VII

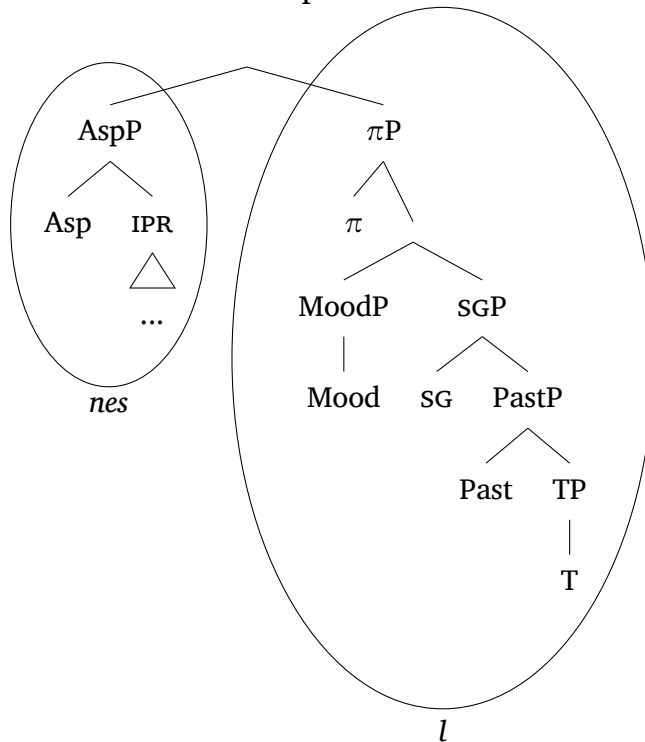
a. Merge F b.

Evacuate closest



In this tree, π P is perfectly identical to the marker *-l*, see (79).¹³

(79) The derivation of the past tense VIII



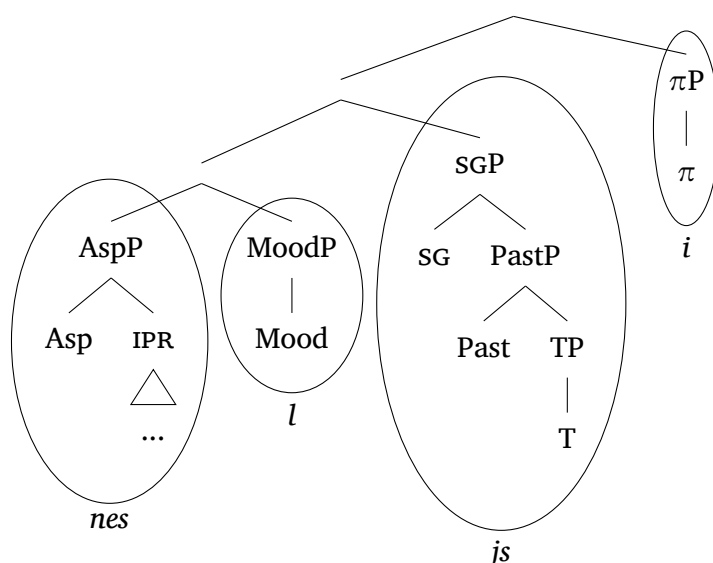
¹³In Nanosyntax, single branching unlabelled nodes (like the one above Mood in (78)) are pruned. I keep it in (78) to keep track of where AspP moved from, but I delete it in (79) for reasons of parsimony.

This is the correct form of the 3RD person singular, with the *-l* lexicalising a large constituent containing its core feature, Mood, but also the feature that can also be lexicalised as a root of auxiliary, and π . The analysis thus turns the 3RD person singular past (with no null auxiliary) into a form fully syncretic with the past participle (lexicalising only Mood).

To get the second person singular, we need to add the PART feature on top of (79). When this happens, no evacuation movement produces a lexicalisable outcome. Therefore, it is impossible to merge PART on top of (79) and get any lexicalisable outcome.

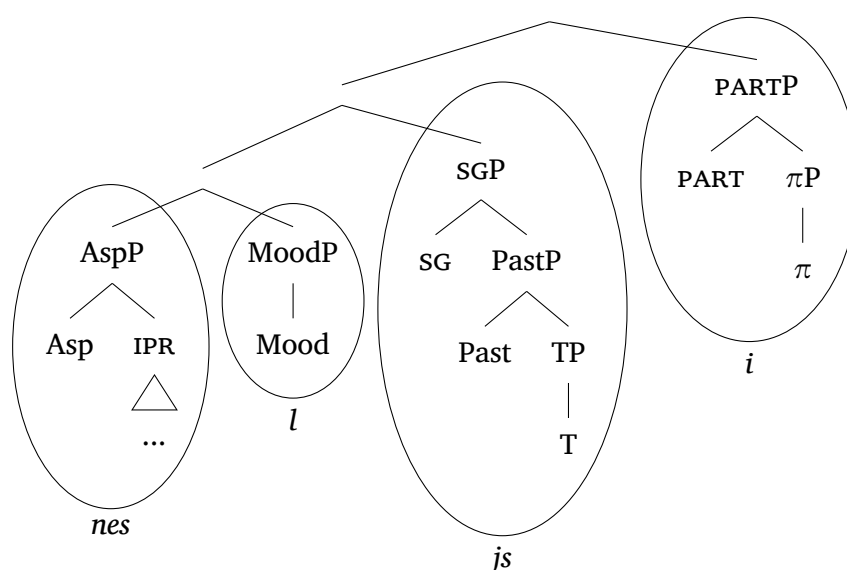
Therefore, the derivation has to backtrack into the previous stage where π was added (recall (78)). Recall that when π was added, we could just extract AspP and we got a successful lexicalisation that we see in (79). However, it is impossible to merge the feature PART to this structure and get a lexicalisable outcome, therefore, we need a different structure for π P. To get it, we backtrack to the π P stage in (78), and we try some other lexicalisation option. The next option to try (as defined by the algorithm) is one where not just AspP extracts, but the immediately larger constituent containing Mood extracts, yielding (80). In this tree, π P can be lexicalised by the 2ND person ending *-i*.

(80) The derivation of the past tense IX



Now recall that this is just an intermediate stage on the road to the 2ND person: what we have achieved so far is just an alternative lexicalisation of the π P; the next step is to reintroduce the 2ND person feature PART into the derivation. When this happens, first the AspP extracts, which does not lead to a lexicalisable outcome. Therefore, larger and larger constituents are pied-piped, leading ultimately to (81). Here, the 2ND person *-i* still applies. Based on the list of the lexical entries introduced in (71), this is the correct structure and output for the 2ND person.

(81) The derivation of the past tense X



The result is non-trivial: what we achieved is that when a *single feature* is added to the 3RD person (as given in (79)), *two new morphemes* appear (*js* and *-i*). This may appear paradoxical for a strictly agglutinative theory of affixation, where the addition of a single property (PART) should lead to the appearance of a single morpheme: but we see two. In the current system, deriving this pattern requires the combination of backtracking and lexical items with a complex left branch.

Obviously, a derivation that would not use these devices would be simpler, and we can expect that there are going to be pressures towards the elimination of such a case. The point is that this is precisely the direction in which Polish developed, recall the paradigm (82). What we see here is that Polish completely lost the auxiliary root *js* throughout the singular paradigm, and it adds the 2ND person marker directly to the participle 3RD person form, avoiding backtracking, and allowing for simpler entries.

(82) Past tense in three varieties of Slavic

	Old Czech	Literary Czech	Polish
1	nes-l js-em	nes-l js-em	nios-ł-em
2	nes-l js-i	nes-l js-i	nios-ł-eś
3	nes-l jes-t	nes-l	nios-ł

However, while the Old Czech and the Polish systems appear uniform and represent a clear source and a clear output of a diachronic change from periphrastic to synthetic marking, it is also important that intermediate systems (like Czech) can be captured. The fact that the current account can generate such intermediate paradigms (as the current section demonstrates) is clearly a good thing.

I close this section by noting that we have seen how the lexicalisation of the 3RD and 2ND person proceeds. I trust the reader can fill in the derivation of the 1ST person on their own, producing the lexicalisation table sketched at the outset, see (83), repeated from (69).

(83) The lexicalisation table for the singular

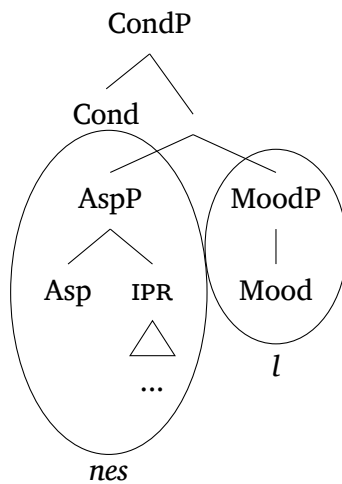
SPI	Asp	Mood	Cond	T	Past	SG	π	PART	SPKR
nes		1				1			
nes		1			js			i	
nes		1			js			em	

5.2 The derivation of the conditional

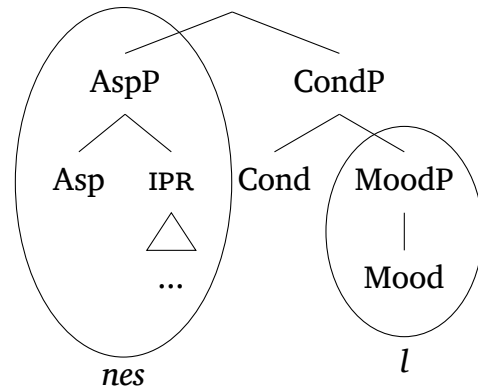
The final section is dedicated to the derivation of the conditional forms. We have already seen in the previous section that MoodP is lexicalised using the verb and the participial *-l* for Mood. We therefore start following the derivation of the conditional at the point when Cond is added to MoodP, as in (84a).

(84) The derivation of the conditional I

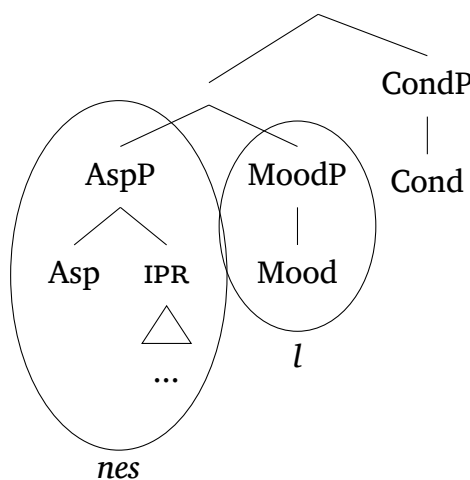
a. Merge F



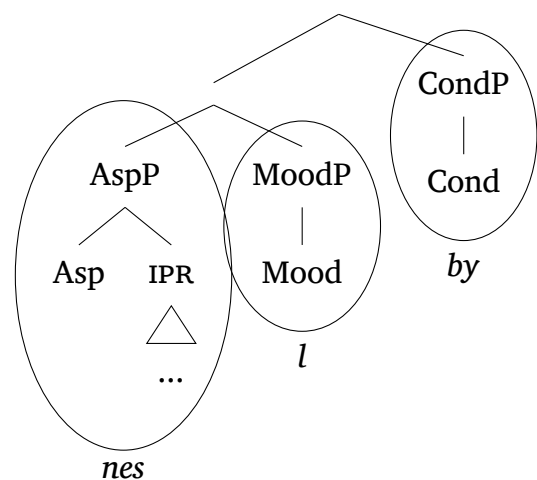
b. Evacuate closest



c. Pied-Pipe



d. Lexicalise

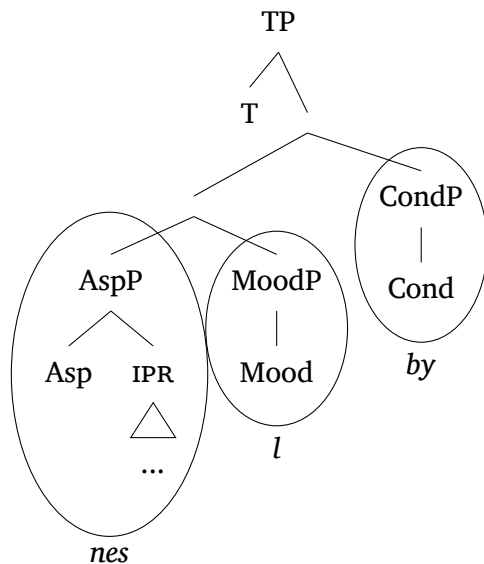


Since there is no match, we try evacuating the closest labelled non-remnant constituent, yielding (84b). But there is no match in (84b) either. We therefore need to undo the evacuation movement of AspP, go back to (84a), and pied-pipe the next higher node. This yields (84c), where Cond is lexicalised by *by*, as in (84d).

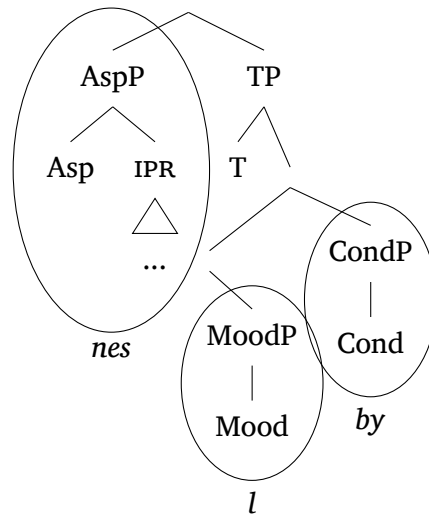
When T is added, as in (85a), there is no match. Since there is no match, we try evacuating the closest labelled non-remnant constituent, yielding (85b). But there is no match in (85b) either. We therefore need to undo the evacuation movement of AspP, go back to (85a), and pied-pipe the next higher node. This yields (85c). The TP in (85c) is matched by *by*, so TP is lexicalised by *by*; see (85d).

(85) The derivation of the conditional II

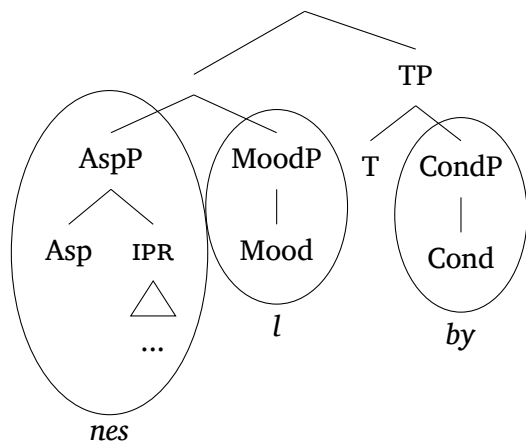
a. Merge F



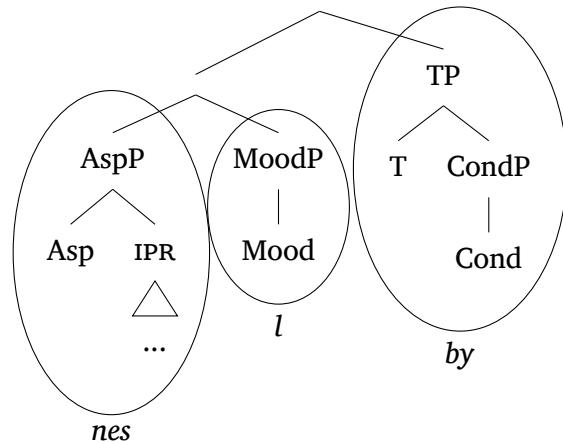
b. Evacuate closest



c. Pied-Pipe



d. Lexicalise

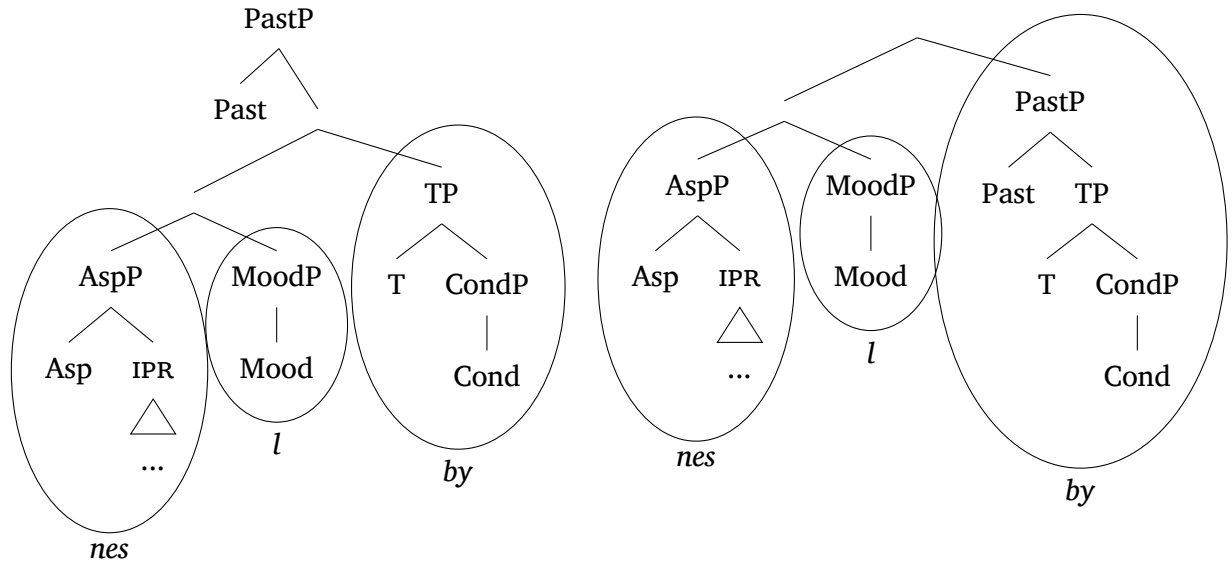


Similar steps apply when Past is merged. I only show the merge structure in (86a) and the final structure in (86b), which is derived by first evacuating AspP, and then pied-piping Mood.

(86) The derivation of the conditional III

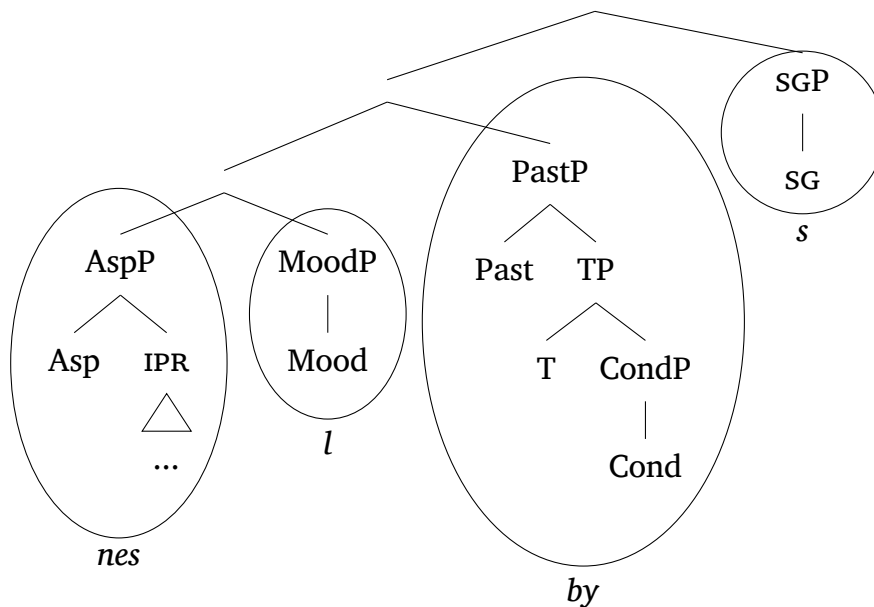
a. Merge F

b. Pied-pipe and Lexicalise



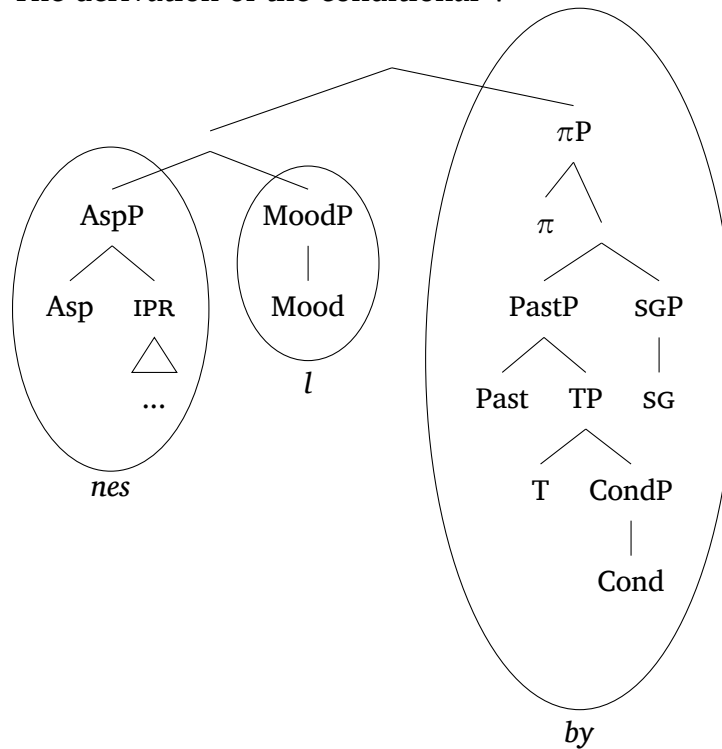
When SG is added on top of (86b), we need to first evacuate AspP, then pied-pipe Mood, but none of this works. Only when we move the whole complement we can lexicalise SGP by -s, see (87).

(87) The derivation of the conditional IV



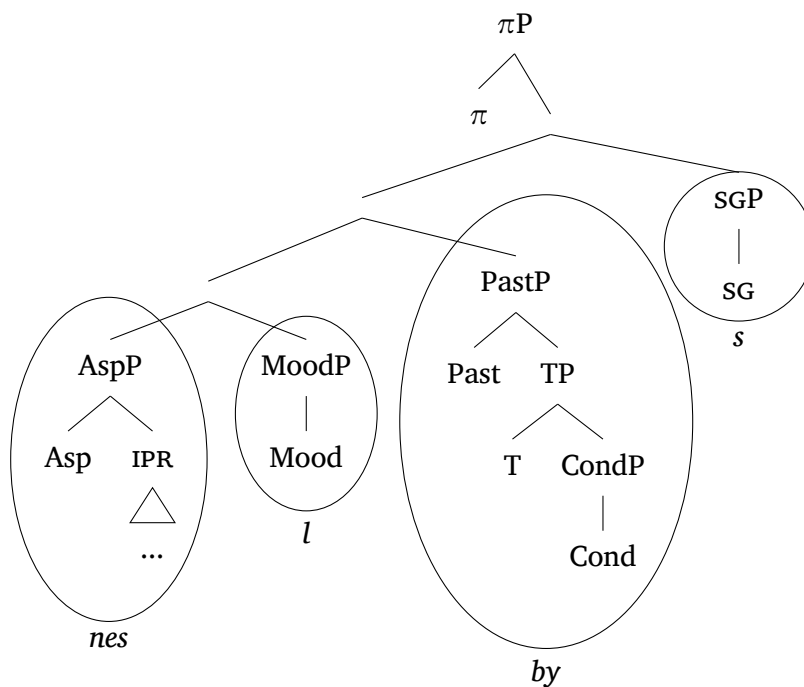
When π is added on top of (87), there is no match. We need to evacuate AspP (no match), pied-pipe MoodP, yielding (88). Here *by* matches the whole π P and we derive the correct form for the 3RD person SG.

(88) The derivation of the conditional V



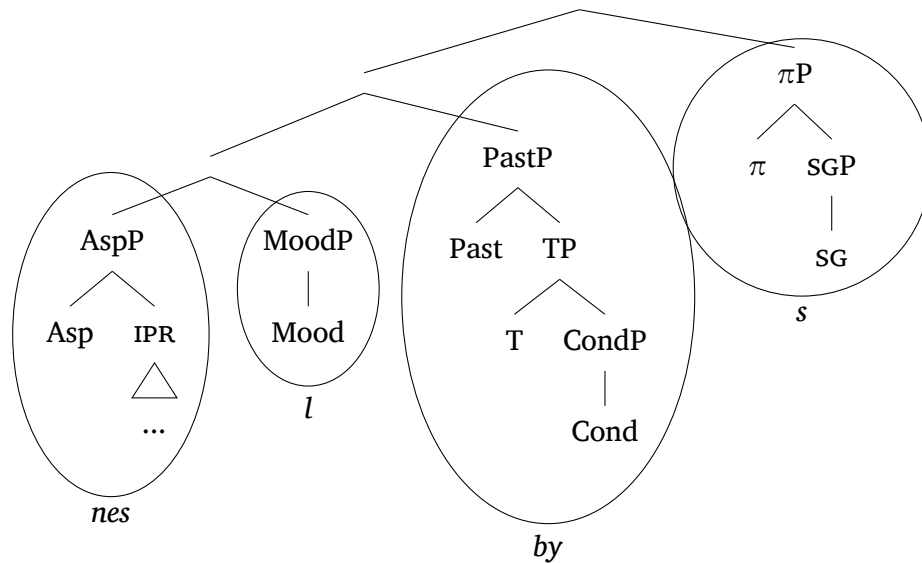
When the 2ND person feature PART is merged on top of (88), nothing works. We therefore need to backtrack to the previous cycle, which is the πP cycle shown in (89). On the first pass, this cycle was lexicalised by evacuating [AspP + MoodP], but now we need to pied-pipe an even higher node, containing also PastP.

(89) The derivation of the conditional VI



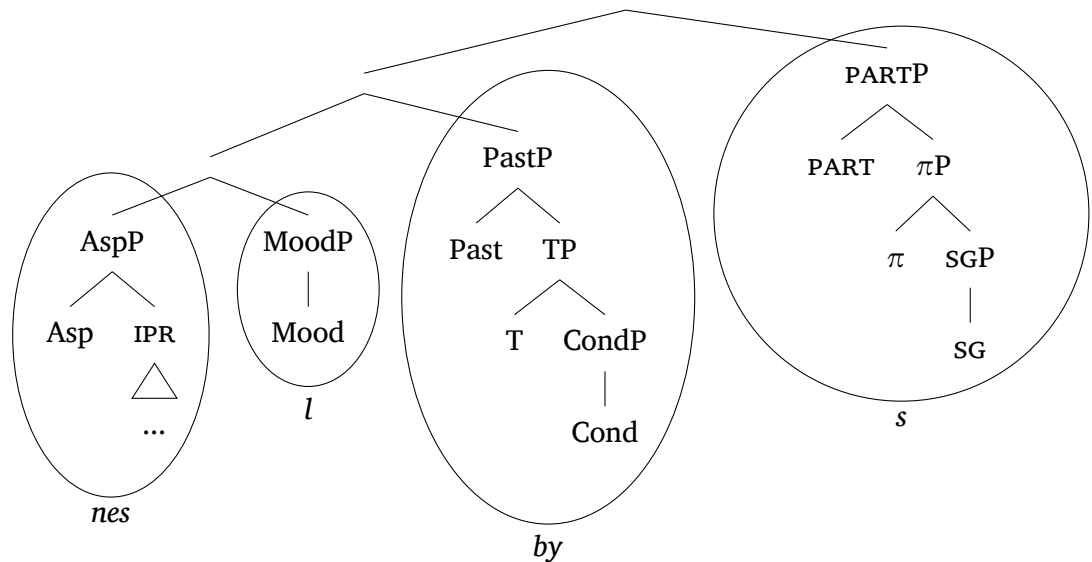
The result is shown in (90), where π P is successfully lexicalised by -s.

(90) The derivation of the conditional VII



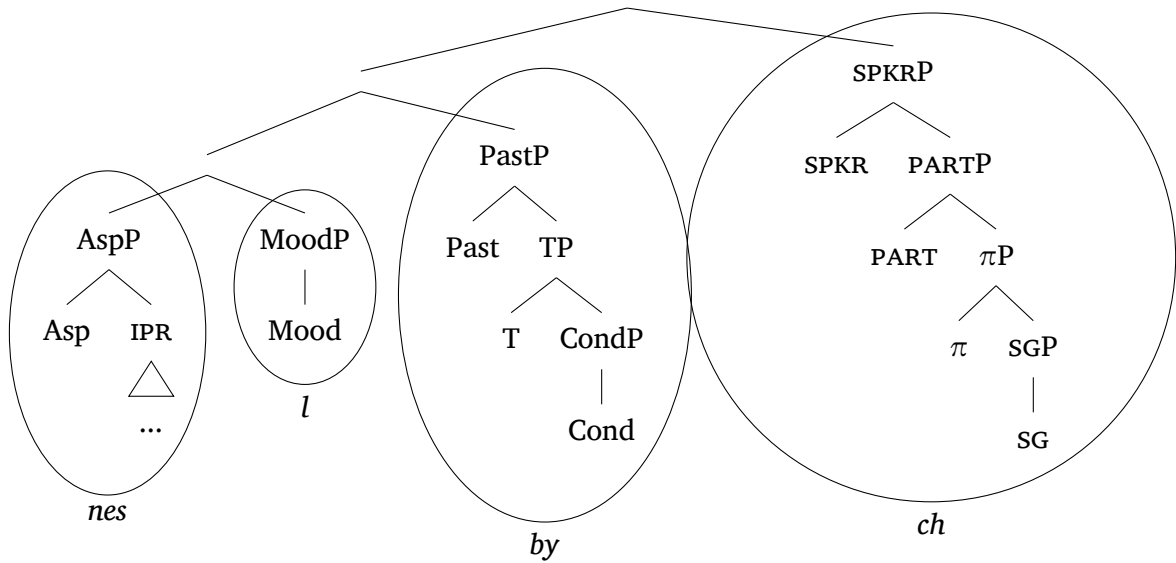
To get the 2ND person, ADDR is merged on top of (90), and evacuations begin. First AspP, then [AspP + MoodP], but these lead to no match. When the next higher node is evacuated, we get the correct 2ND person form, see (91).

(91) The derivation of the conditional VIII



To get the correct form of the first person, we now only need to add SPKR feature to (91). AspP is evacuated (no match), pied-piping MoodP does not help either. Only when PastP is also evacuated, we get the correct form:

(92) The derivation of the conditional IX



The lexicalisations precisely correspond to the lexicalisation table (69), repeated below for convenience.

(93) The lexicalisation table for the singular form of conditional

SPI	Asp	Mood	Cond	T	Past	SG	π	PART	SPKR
nes		l	by						
nes		l	by			s			
nes		l	by			ch			

6 Conclusion

This chapter started from the observation that Czech X-marking conforms to a crosslinguistic pattern of containment whereby the X-marking construction adds an extra morpheme on top of the past tense morphology (Majdić 2025). However, the morpheme order attested in Czech is unexpected: the apparent past tense marker *-l* is found closer to the root than the conditional *by*, which turns out to be an apparently mirror-violating order when compared to the orders found in Turkish or Mauwake.

Relying on the framework of Nanosyntax (and drawing on the account of mirror-violations by Sergienko 2025), I proposed a solution that is based on the idea that the ‘past tense’ *-l* is not just a past tense marker, but a portmanteau that also reaches down to Mood. Using this idea, the apparent mirror-violation can be resolved if the morpheme *-l* does not lexicalise Tense in the problematic constructions, but only a head lower than the conditional (namely Mood). I then showed that this idea is compatible with the full singular paradigm, i.e., taking into account all the person markers in the singular.

The empirical picture is further muddled by the fact that some of the morphemes are second position clitics. The Nanosyntactic implementation I proposed treats the relevant markers as essentially post-markers (markers following the verb). To account for their second position placement, I was led to explore

the idea that in Czech, it's not only pronominal clitics that move to the second position, but also verbal clitics do; i.e., the proposal is that verbal clitics always follow the verb, but when the verb is not first, they have to detach from it and move higher up in the structure. Clitic doubling constructions were offered as one piece of evidence, the interaction between negation and cliticisation as another one.

Future research will explore the implications of this proposal for the interaction of verbal and pronominal clitics, and try to understand the causes and mechanics of language change, where in Polish, the periphrastic past tense is turning into a synthetic form, following the elimination of the auxiliary's root. Finally, the paradigm levelling that is currently taking place in Czech (where the past tense and the conditional unify their auxiliaries) would also be worth a closer look.

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